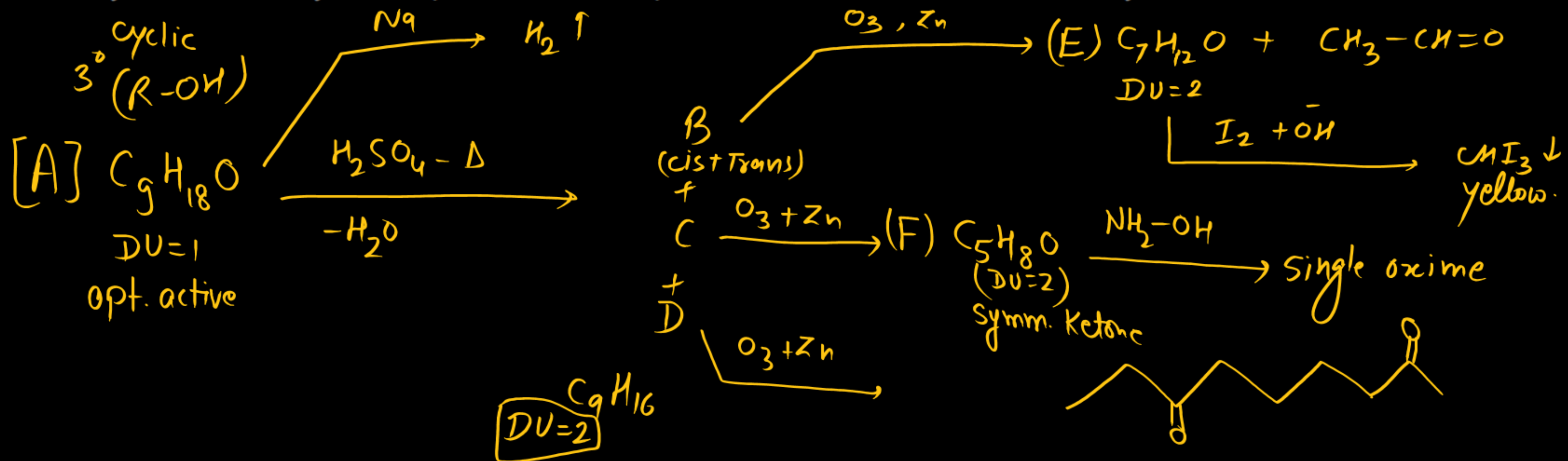
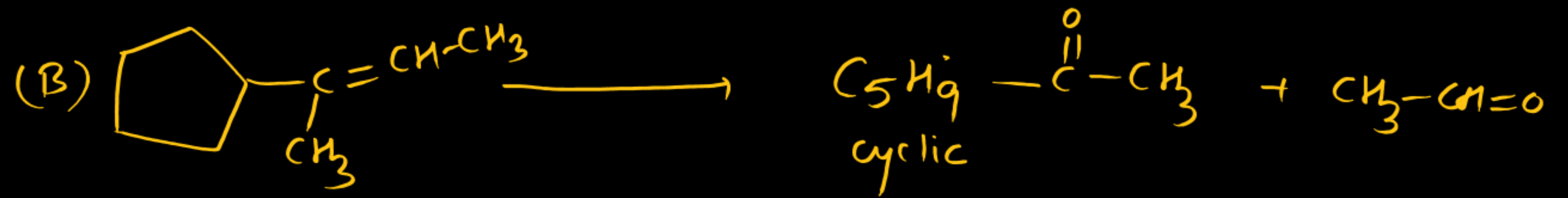


CATS-PLUS

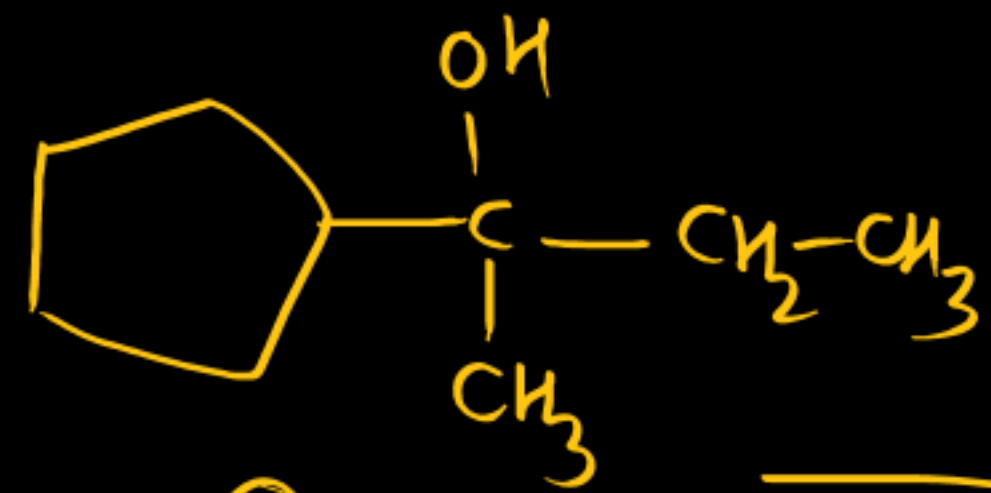
2022

An organic compound A has molecular formula $C_9H_{18}O$ and it can be resolved into enantiomers. A evolves a colourless gas on heating with Na-metal, but does not change the colour of acidified dichromate solution or bromine water solution. A on heating with concentrated sulphuric acid undergo dehydration to yield three isomeric products B, C and D all having molecular formula C_9H_{16} . B shows cis-trans isomerism and on ozonolysis followed by work-up with $Zn - H_2O$ produced E ($C_7H_{12}O$) as one product. E formed a yellow precipitate on treatment with alkaline solution of iodine. Ozonolysis of C followed by work-up with $Zn - H_2O$ produced F (C_5H_8O) which did not produce yellow precipitate with alkaline $NaOH$ but produced a single isomer G (C_5H_9NO) on treatment with H_2NOH . Also D on ozonolysis followed by work-up with $Zn - H_2O$ produced nonan-2,7-dione. Identify A to G.





+

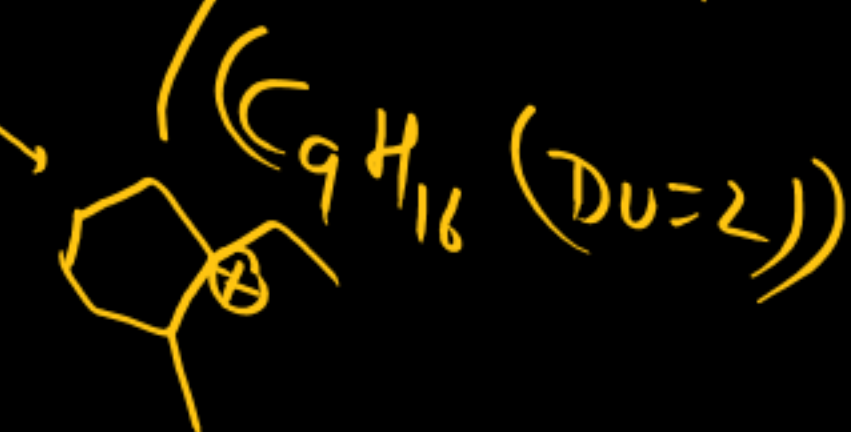
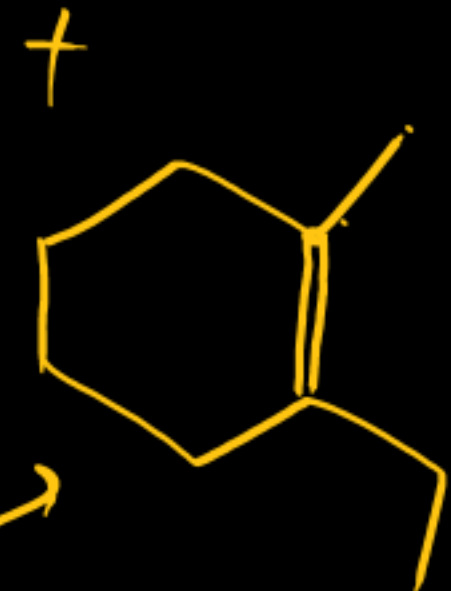


3°
O.Ac
Cyclic

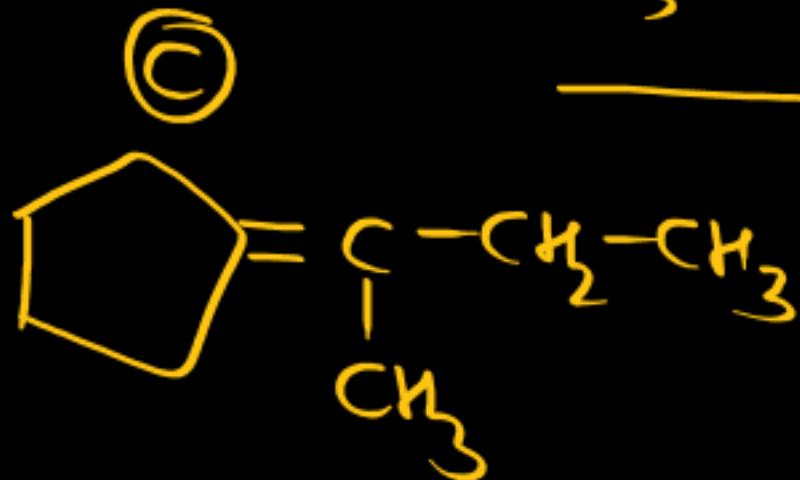
(A)



(D)

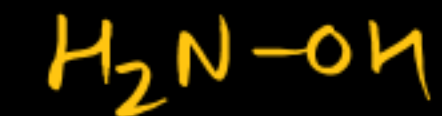


$(C_9H_{16} \text{ (DU=2)})$

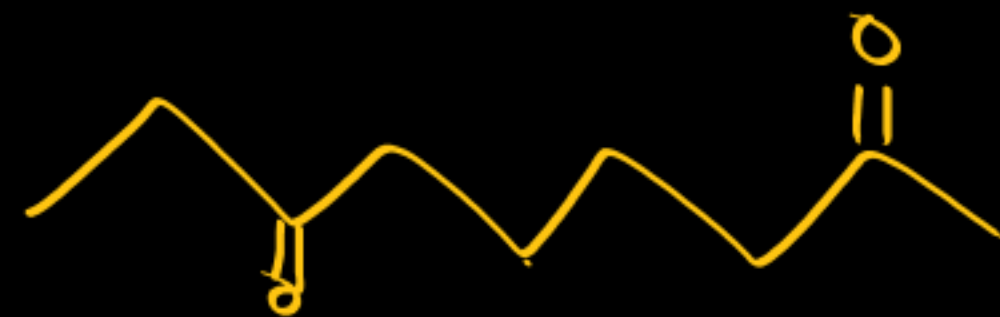


C_5H_8O

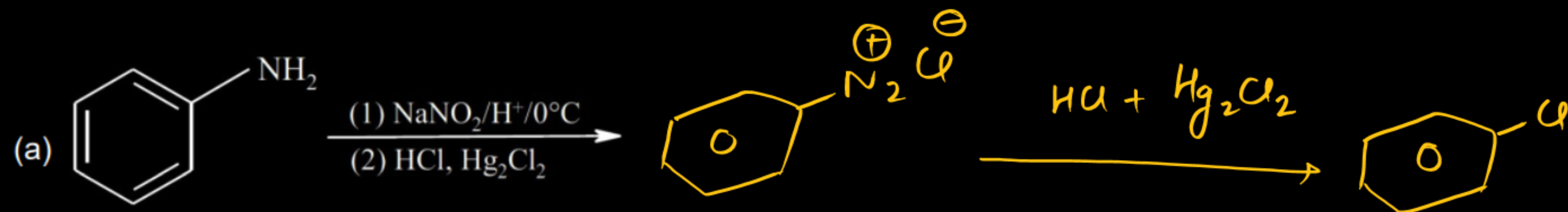
DU=2

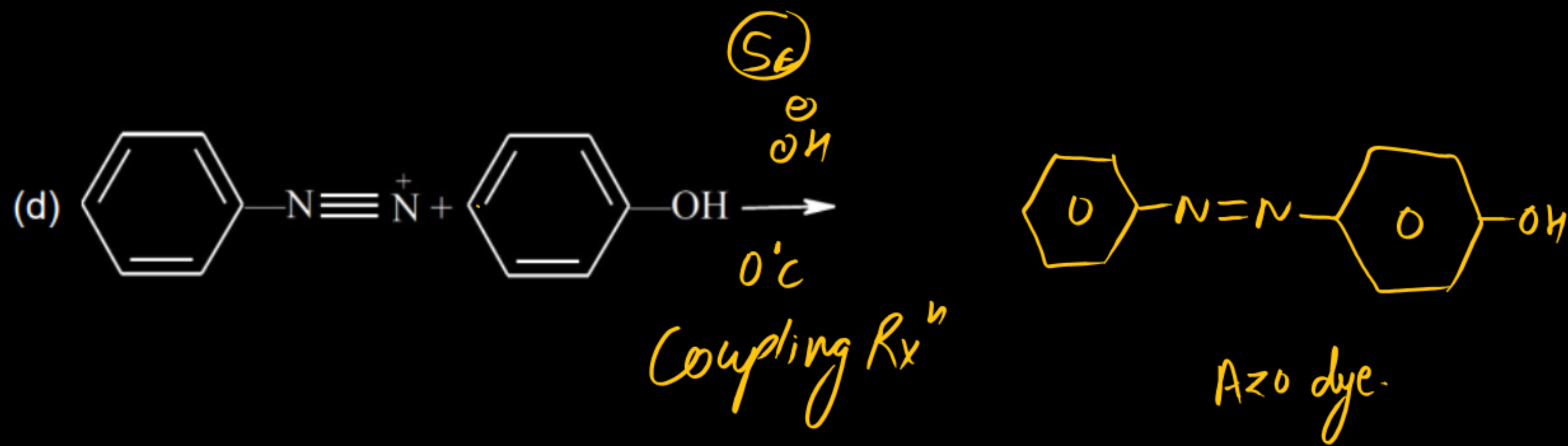
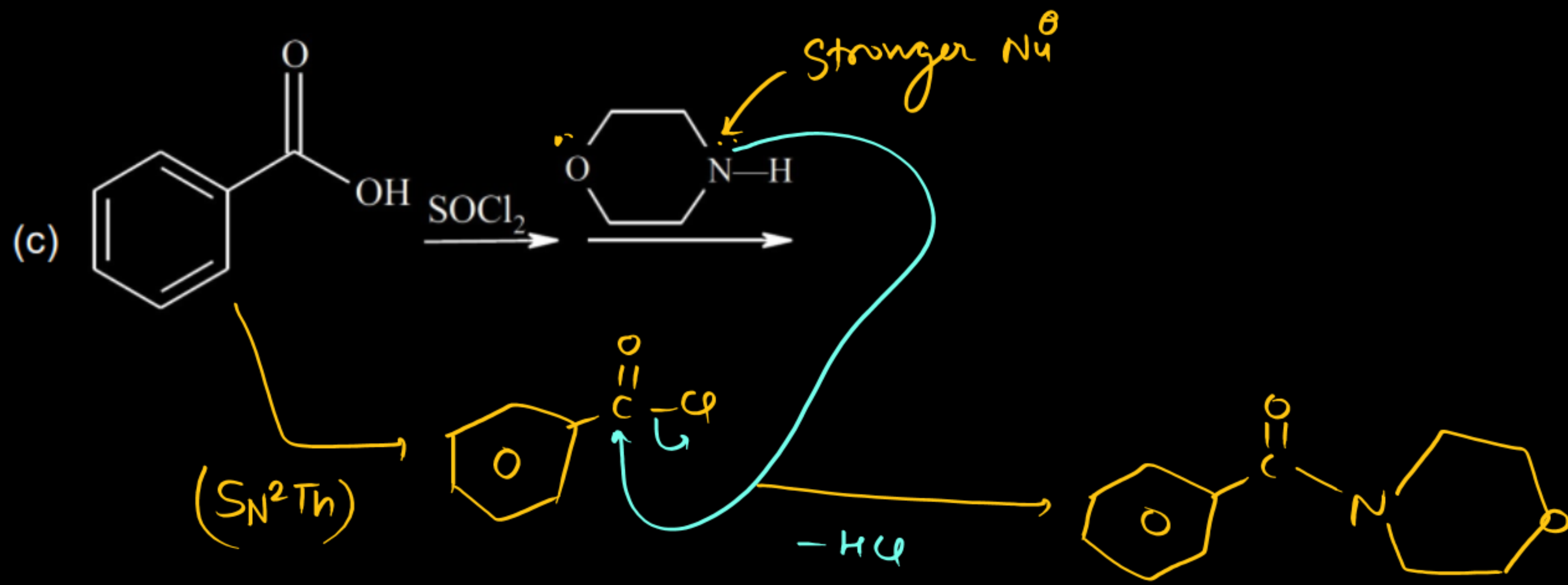


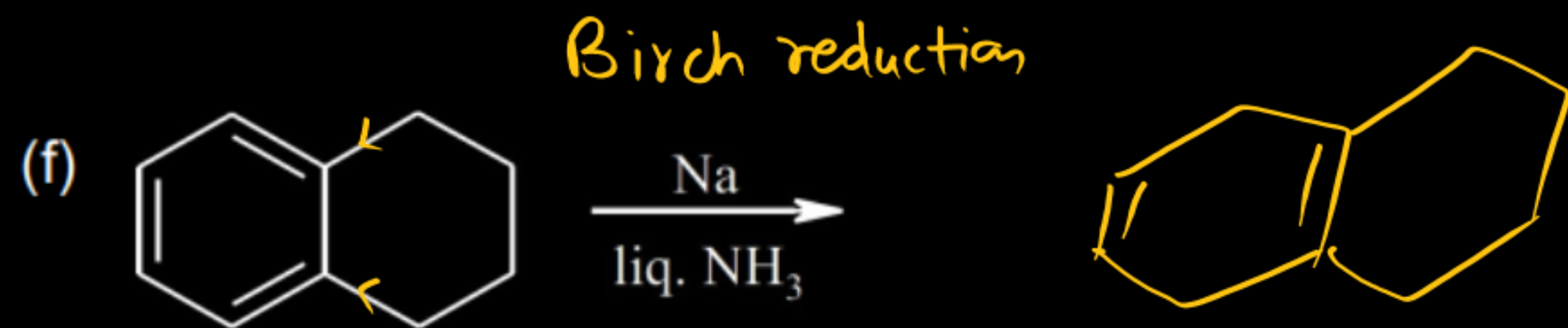
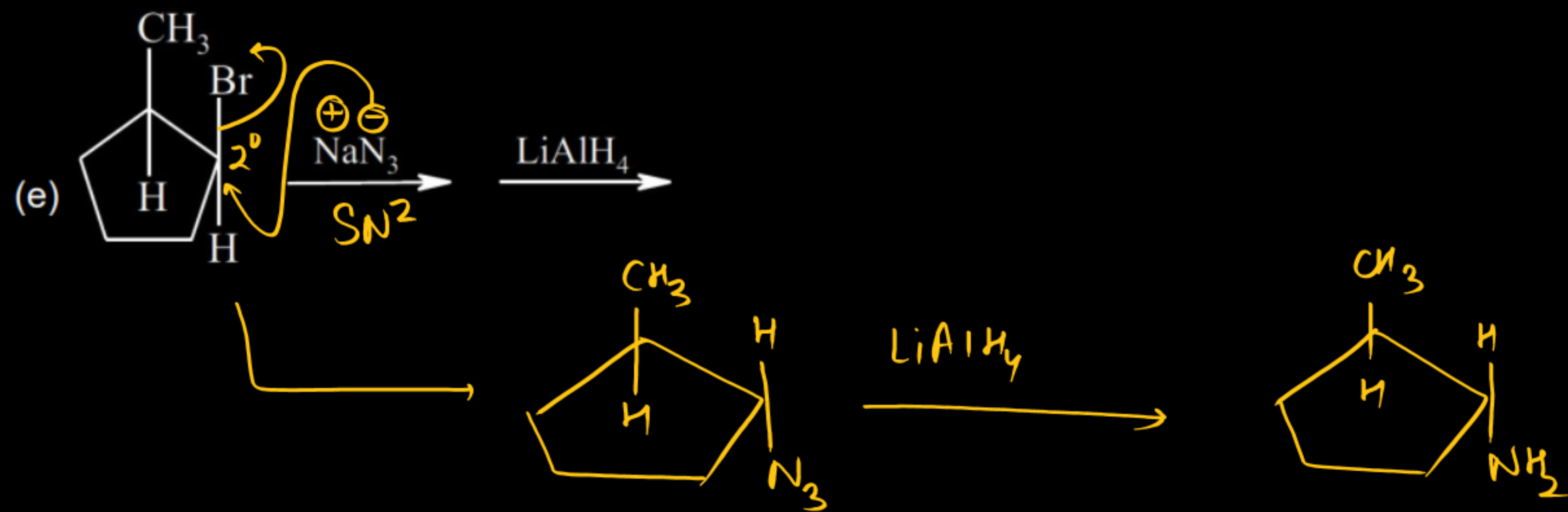
only one oxime

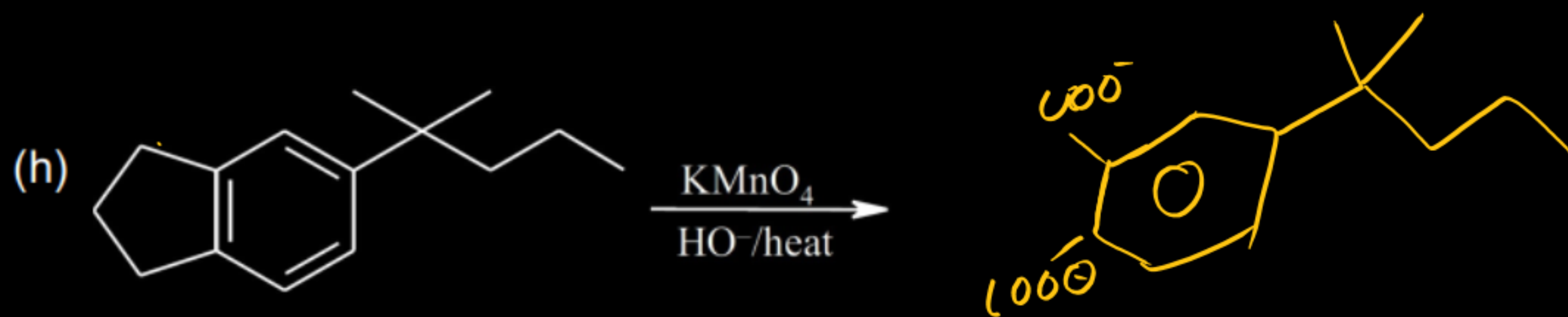
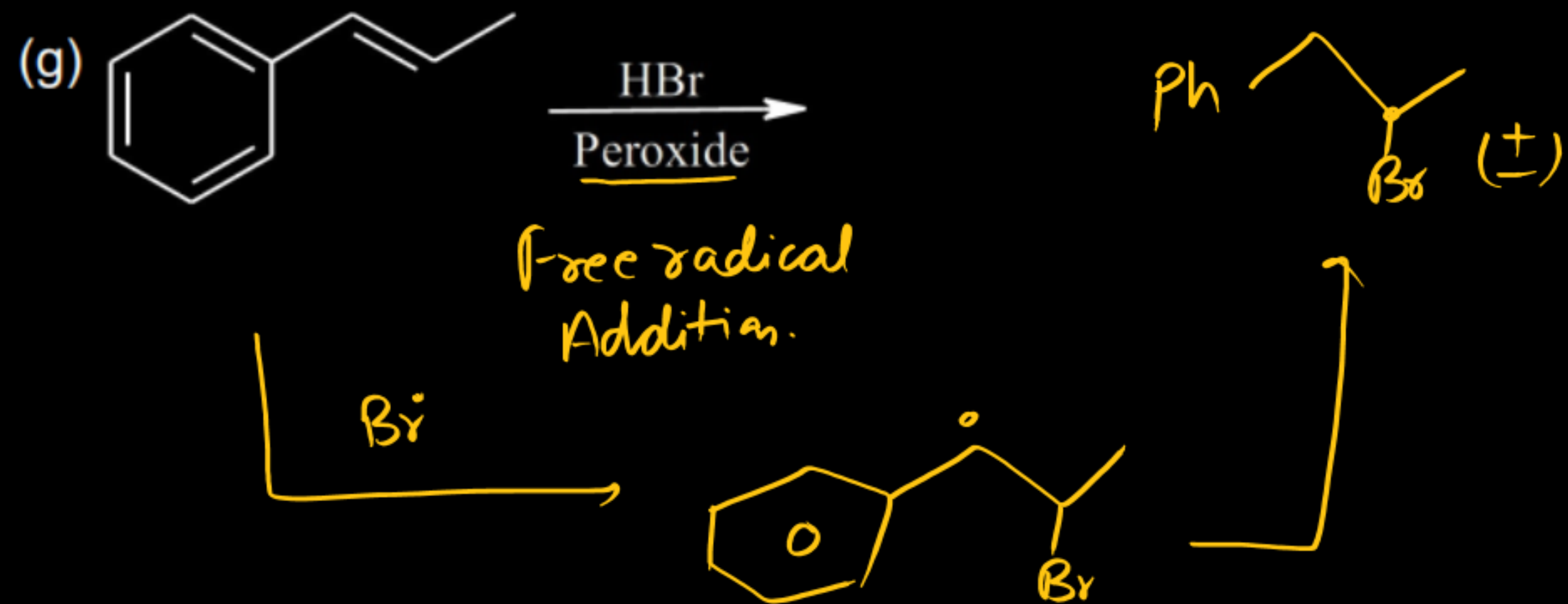


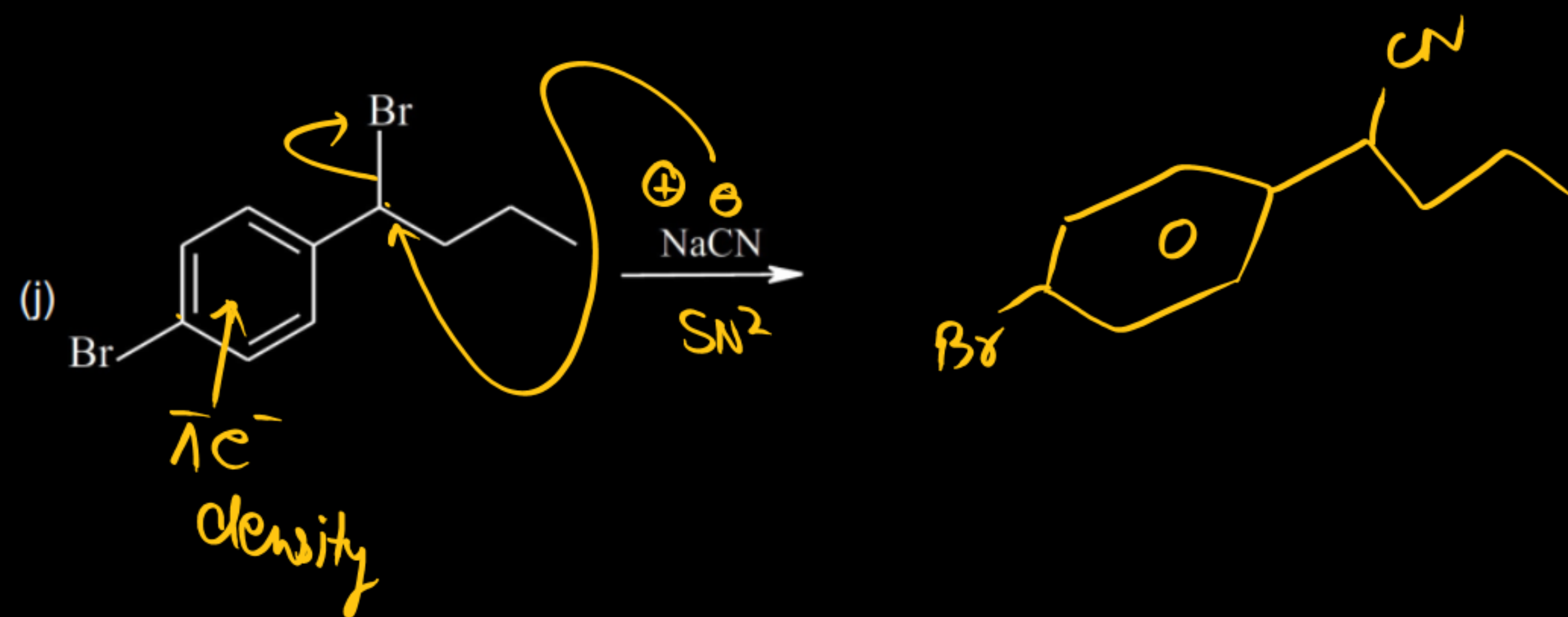
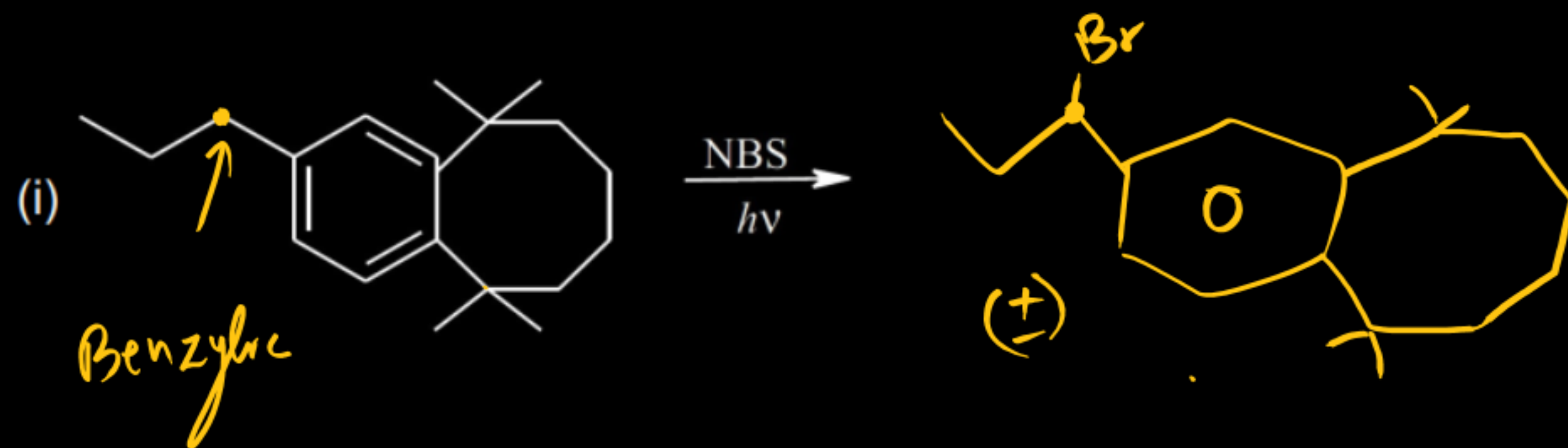
Write major products of the following reactions:

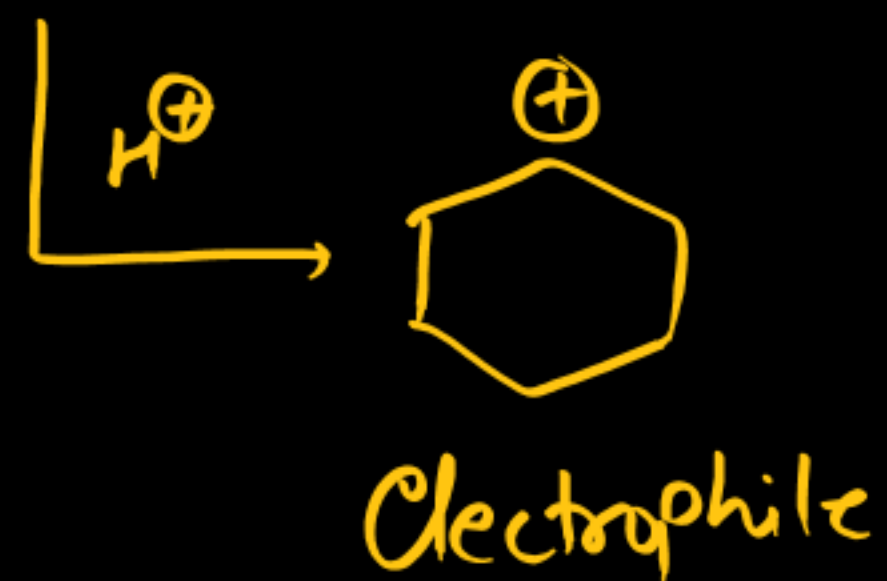




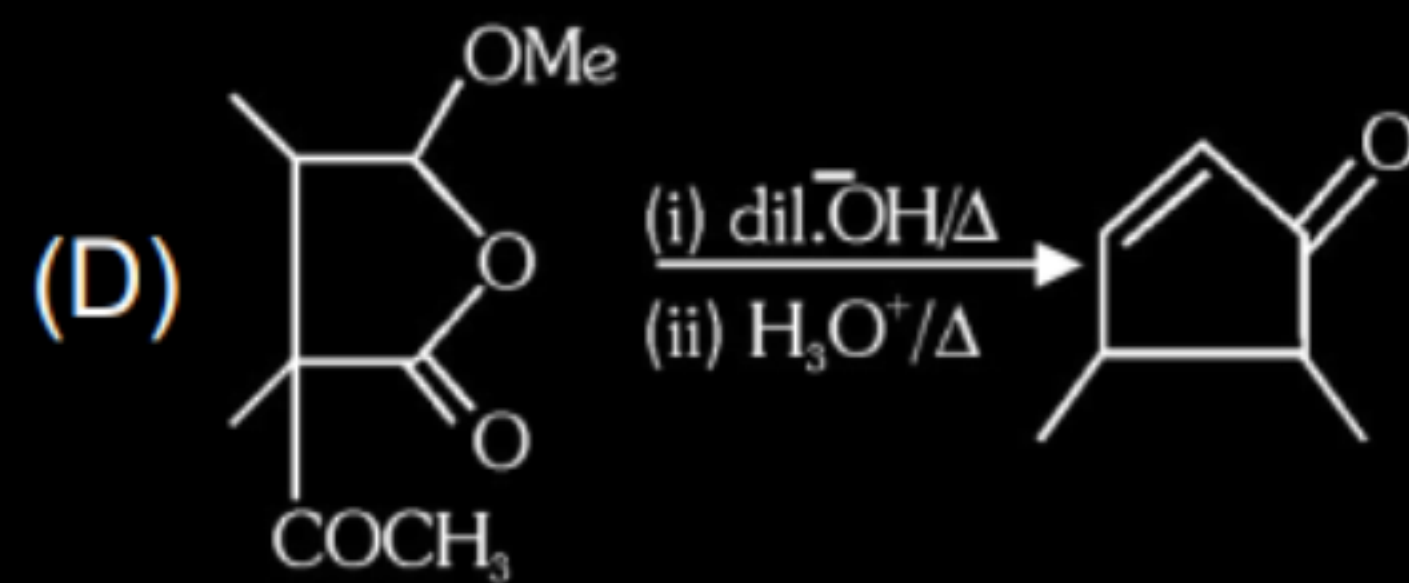
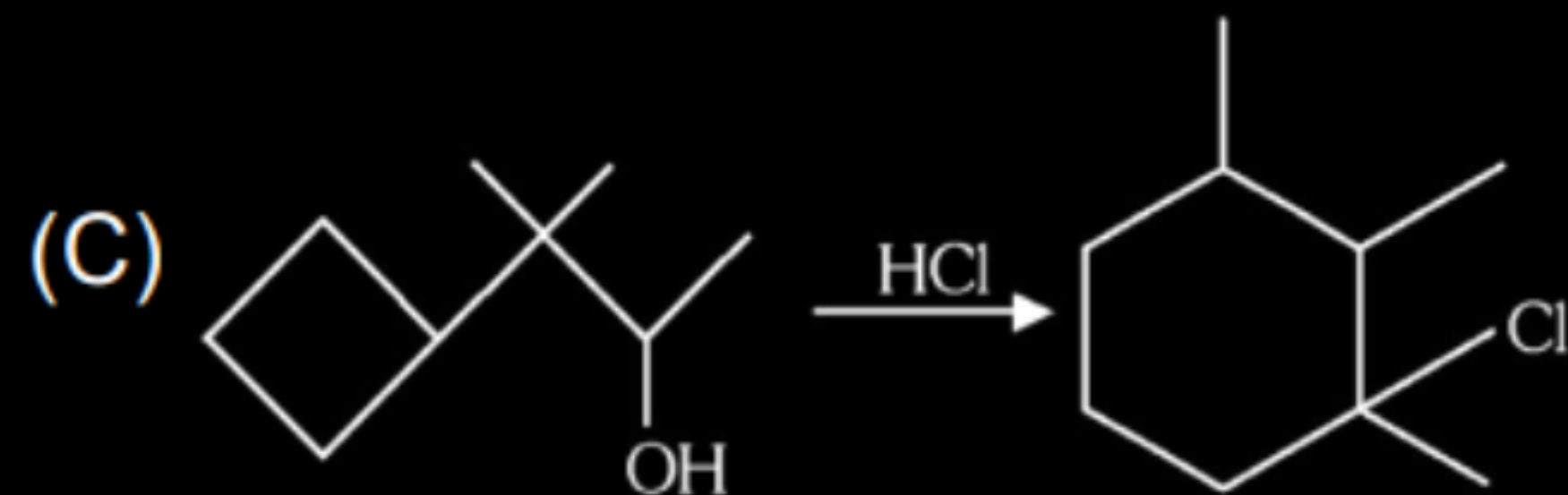
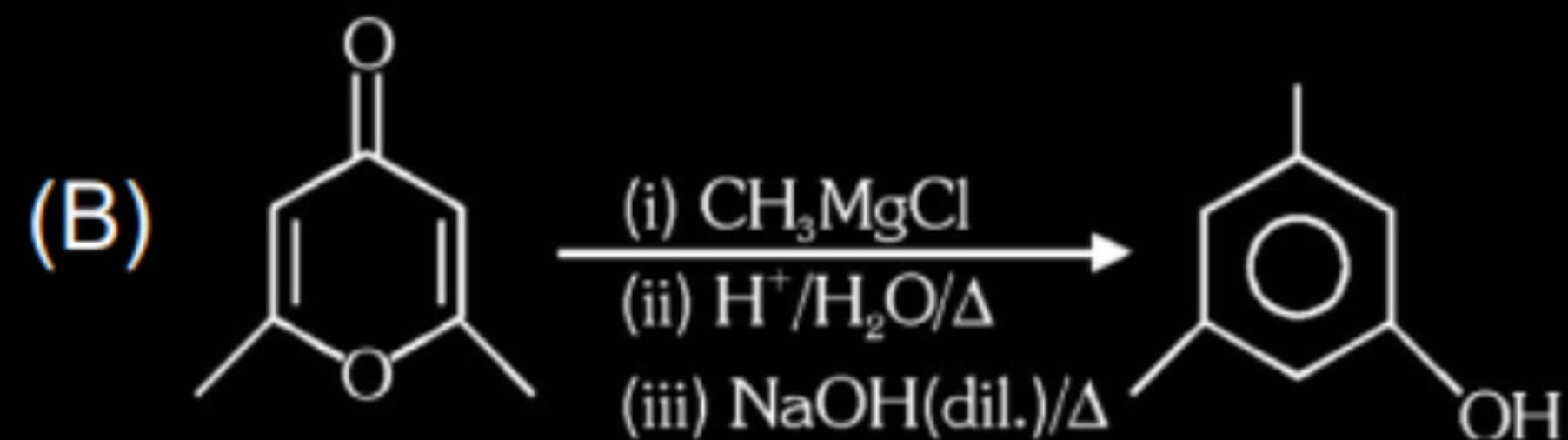
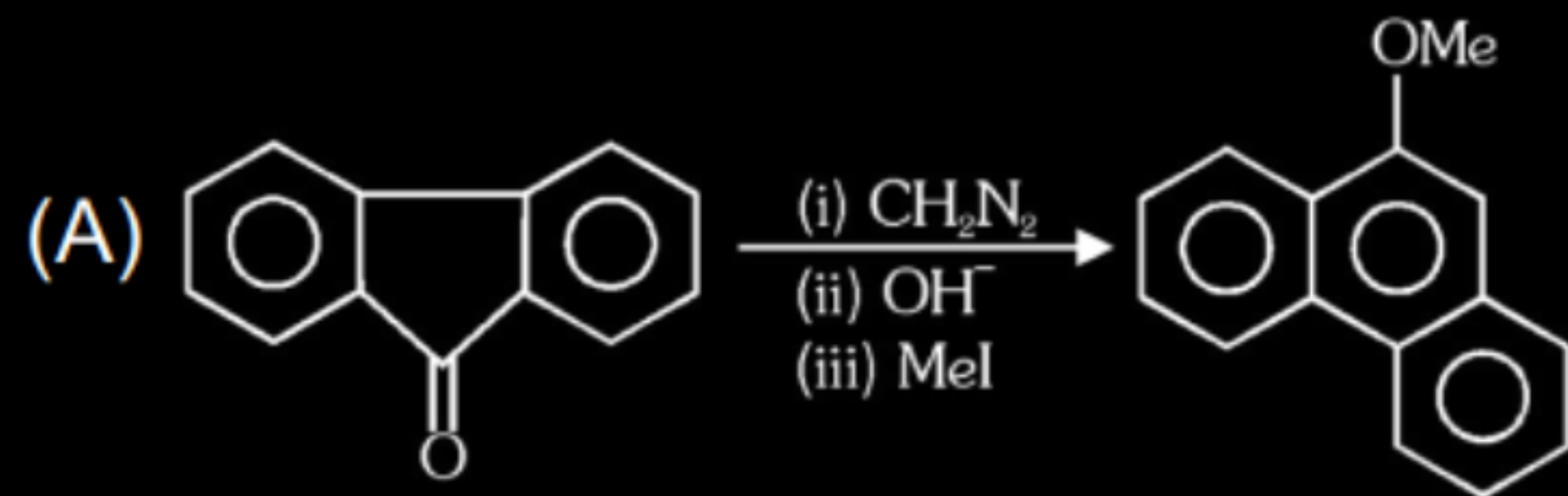


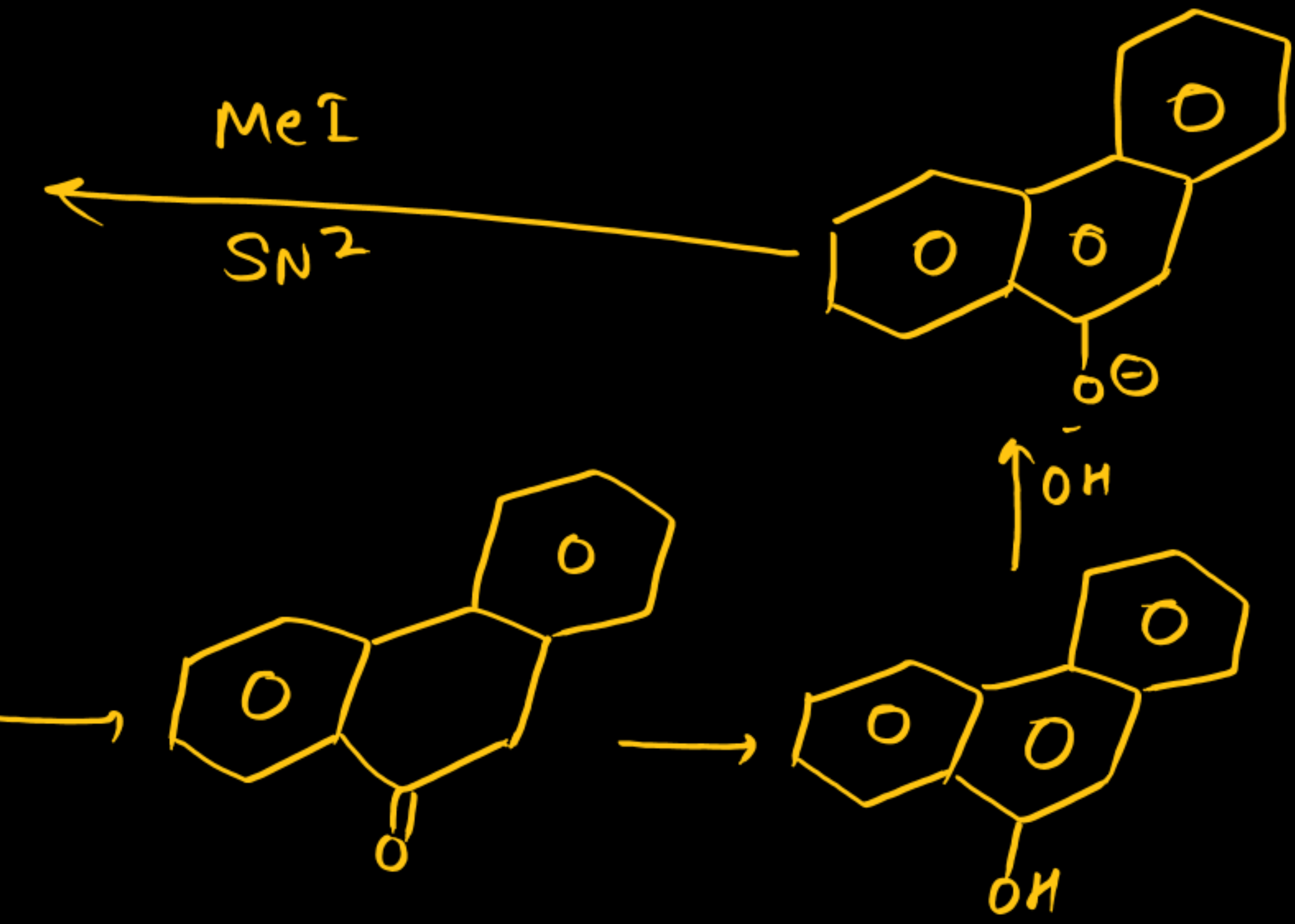
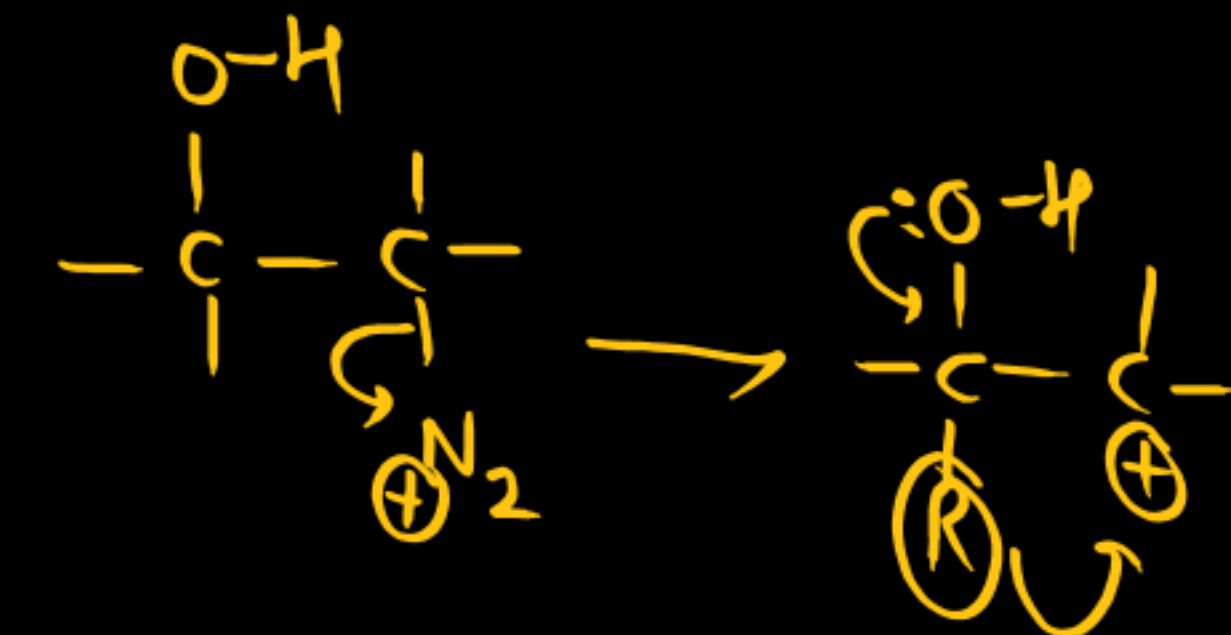
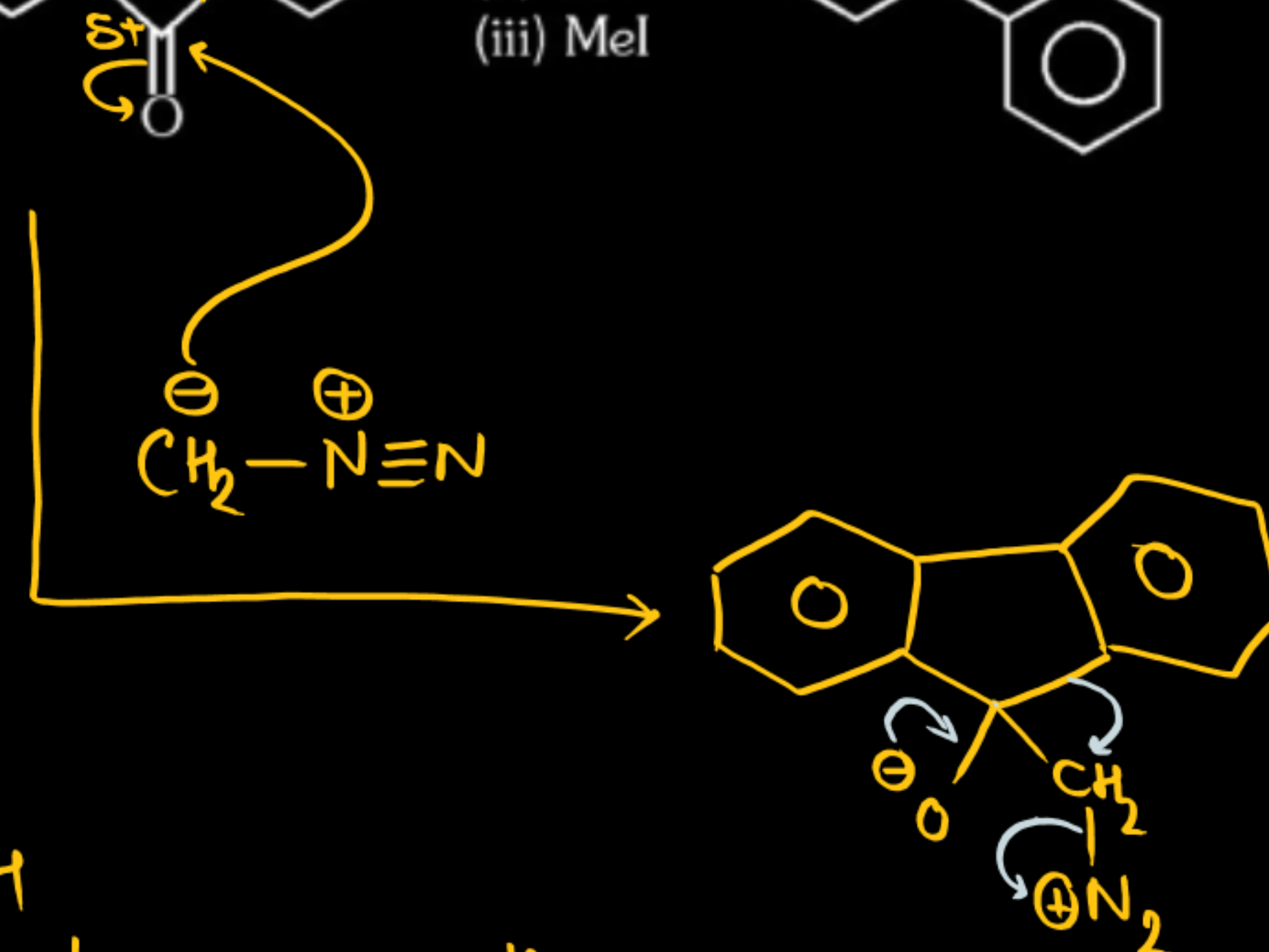
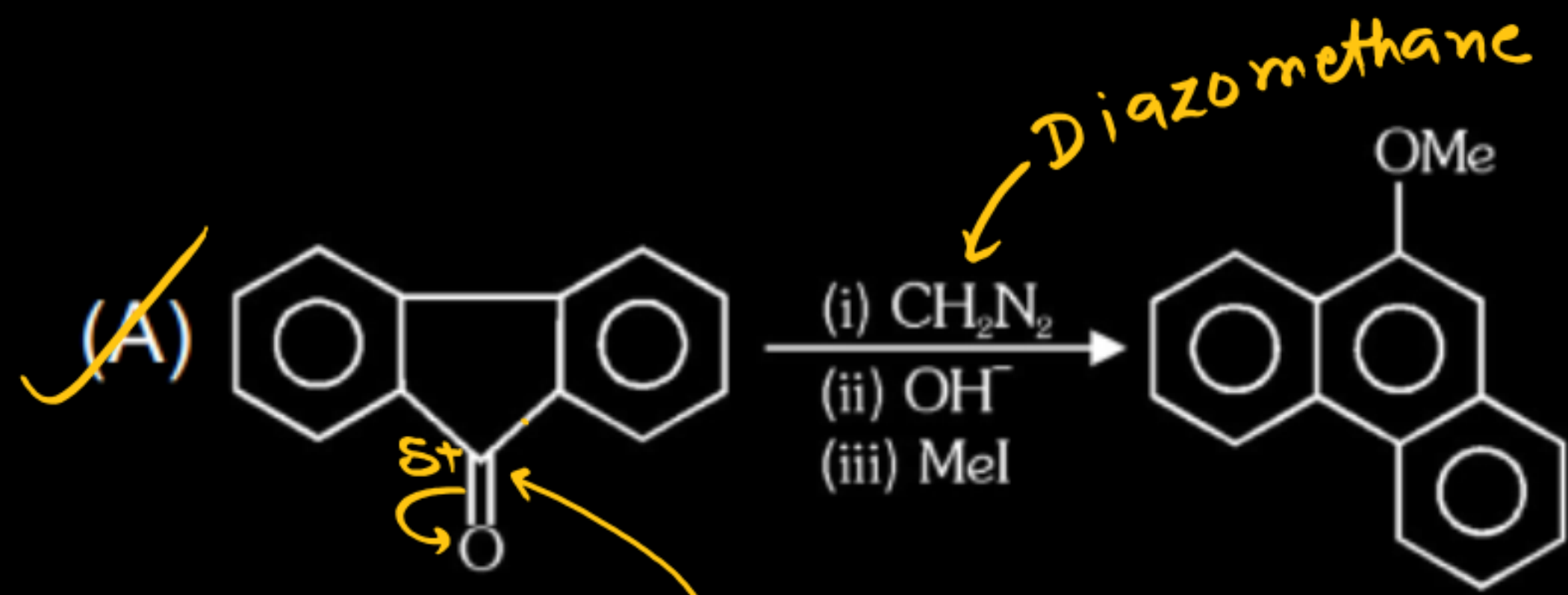




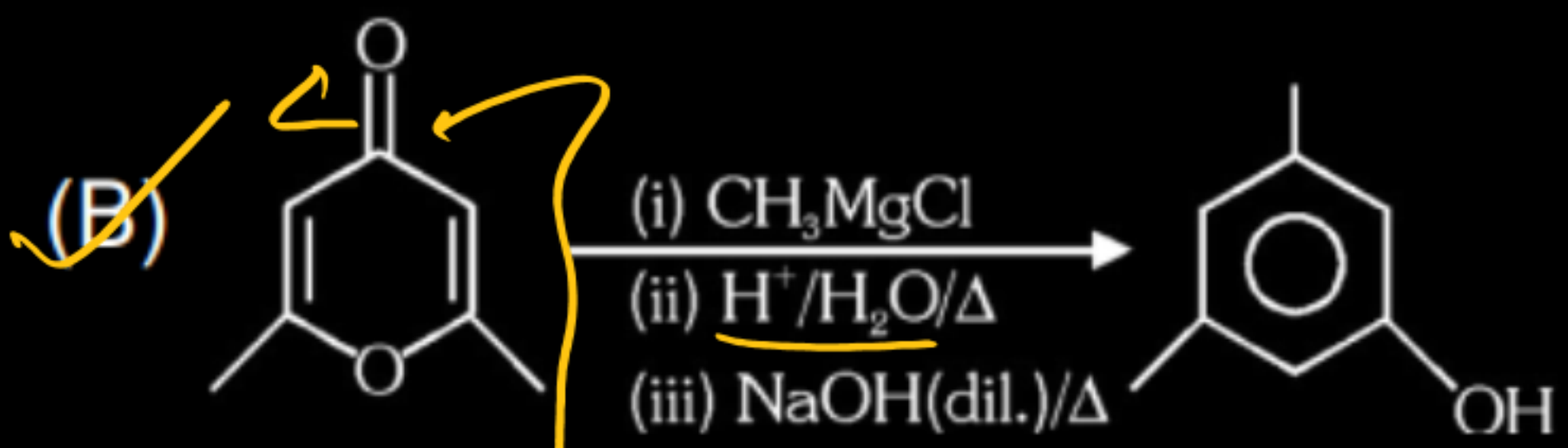
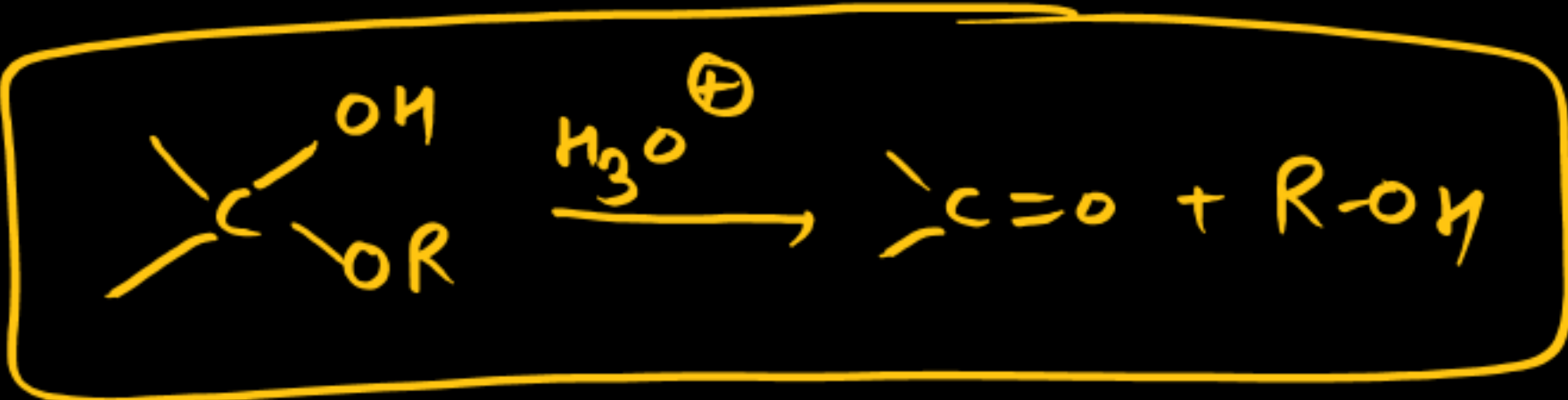


Q Select correct reactions:

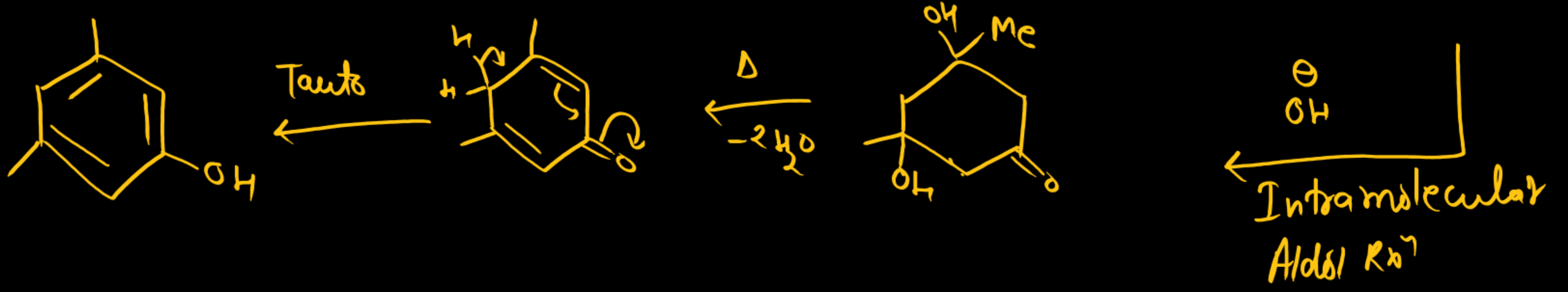
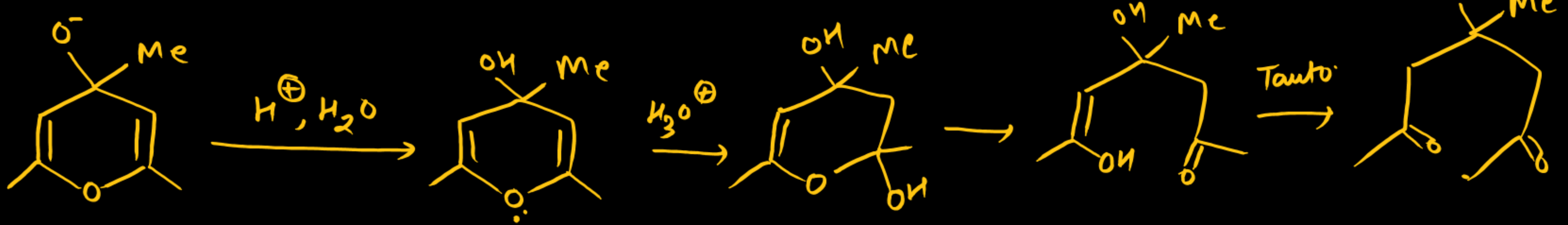


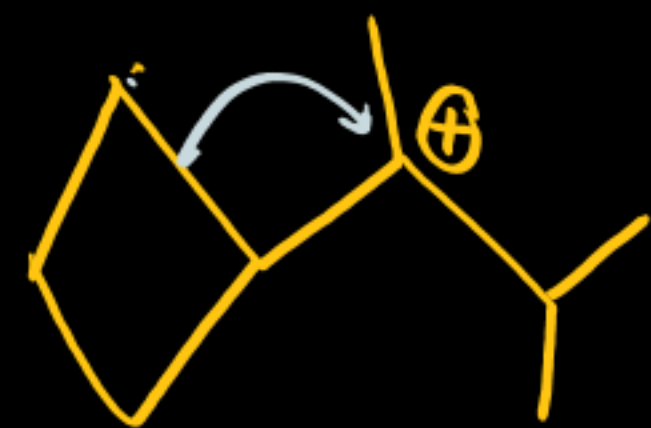
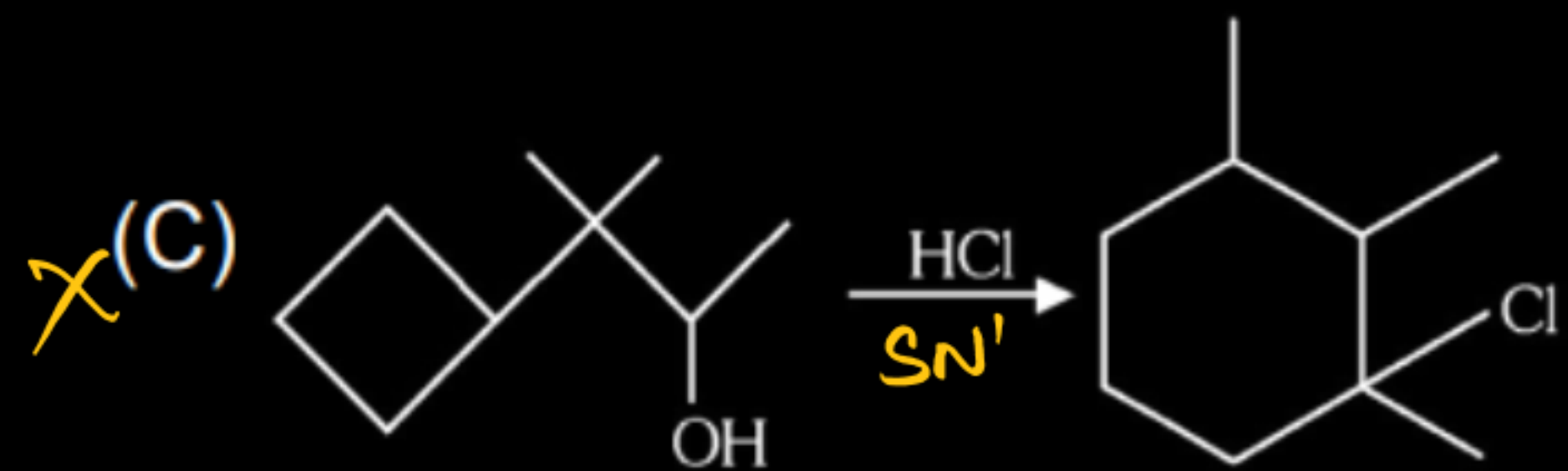


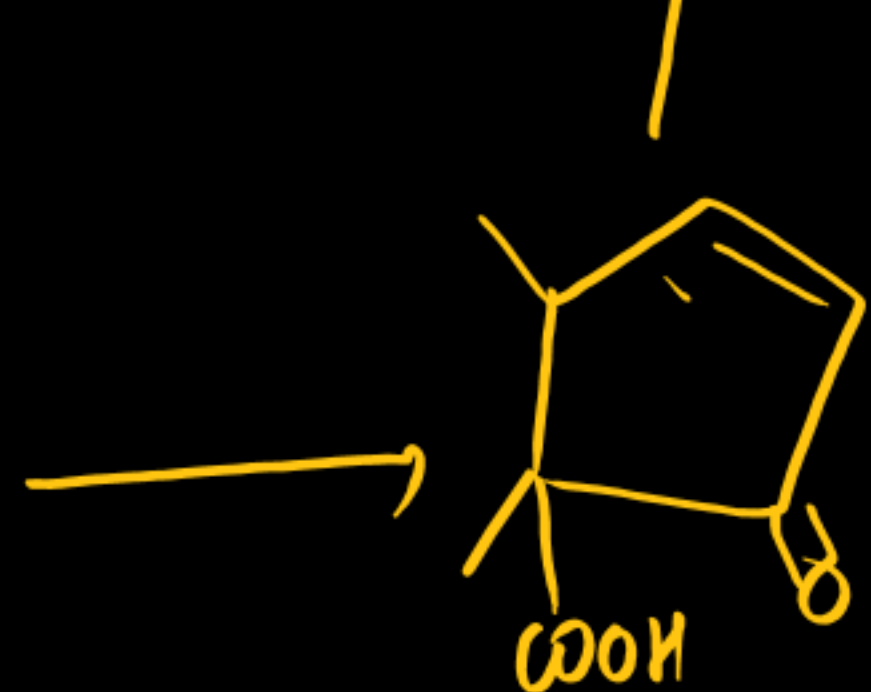
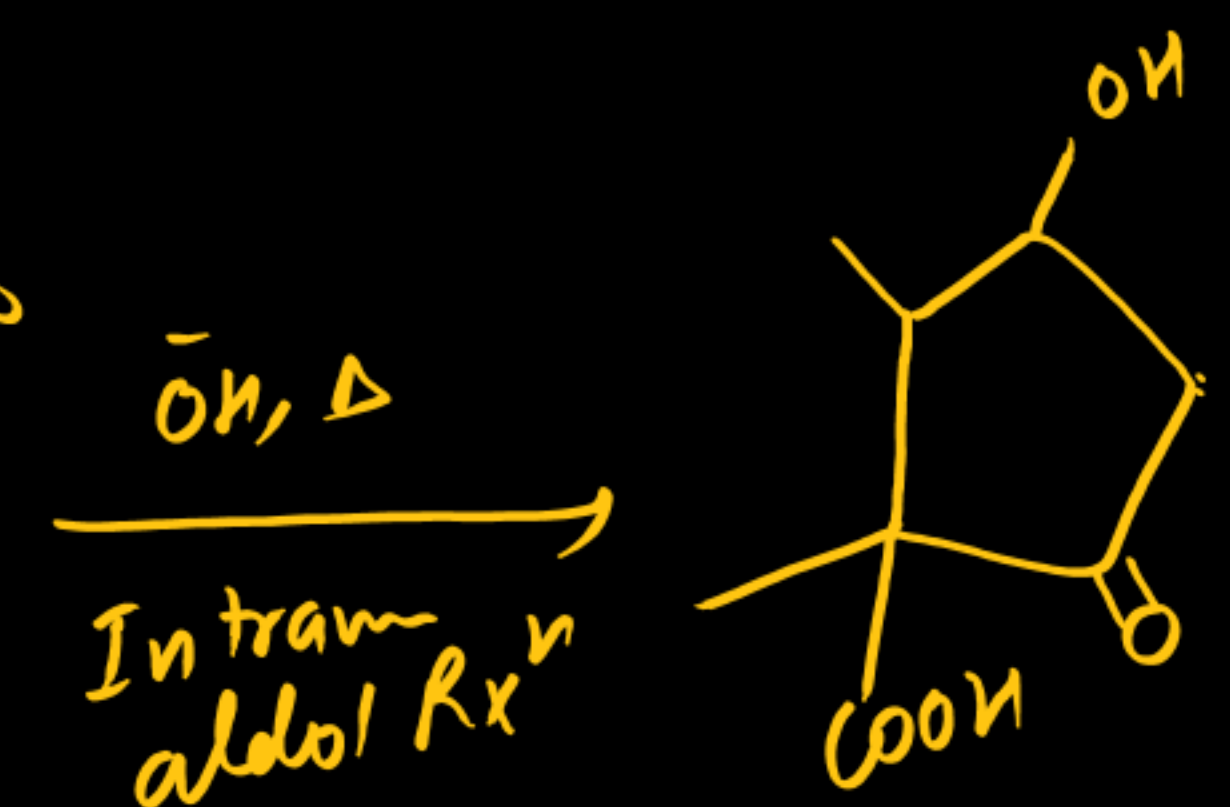
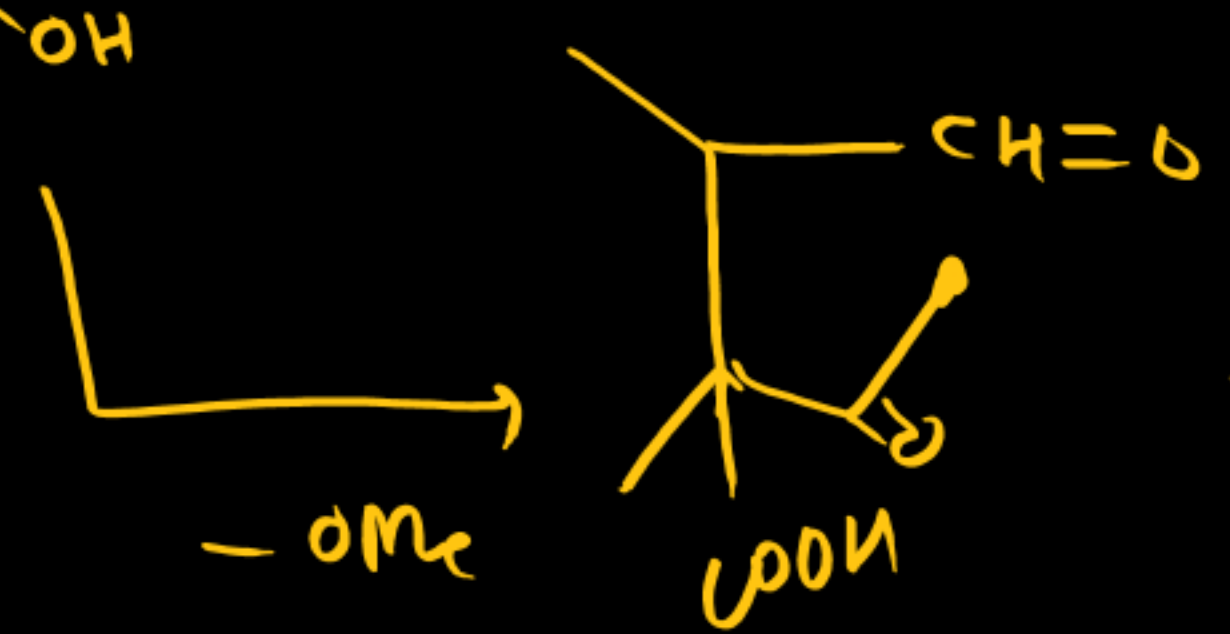
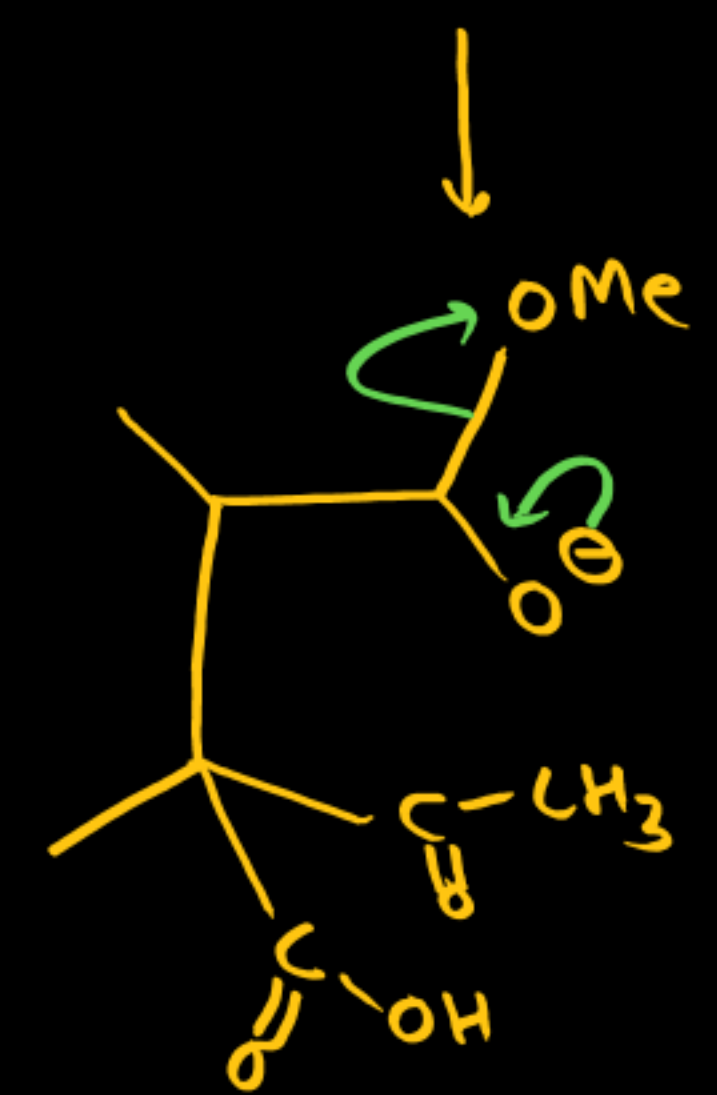
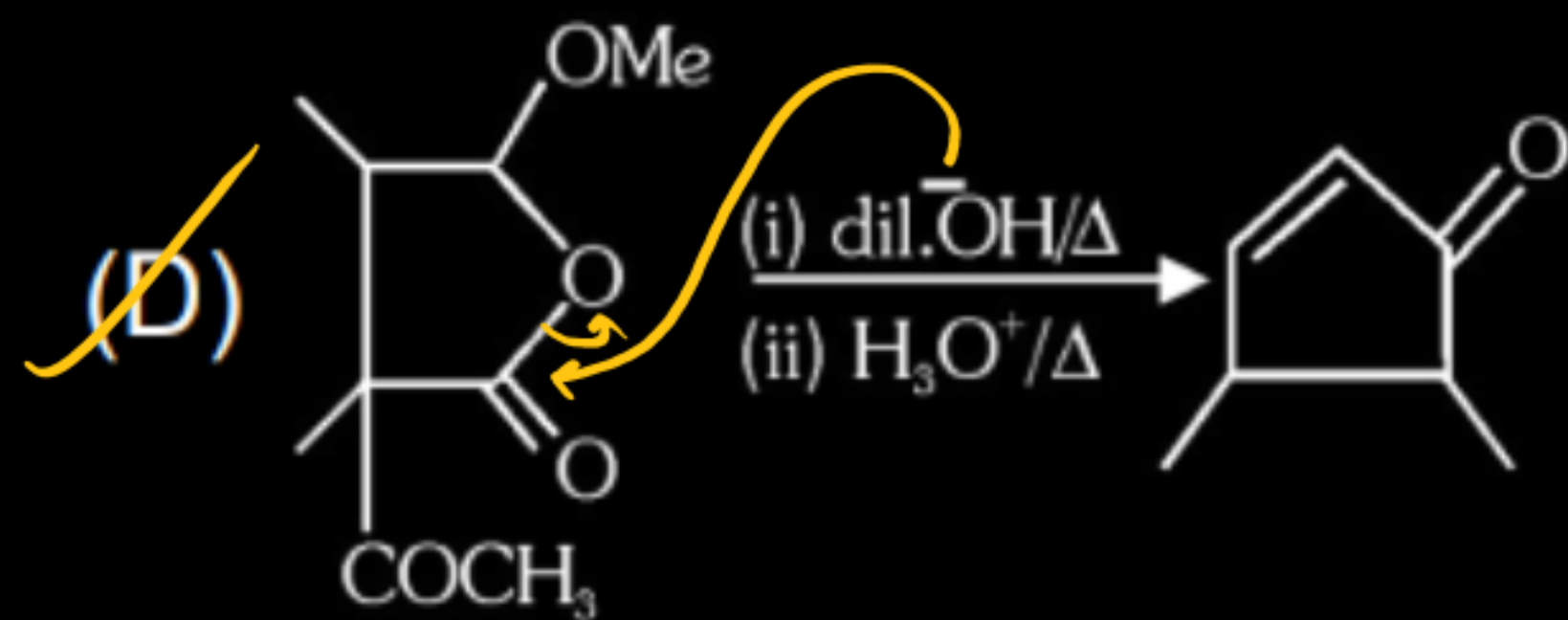
MeI
 $\text{S}_{\text{N}}2$



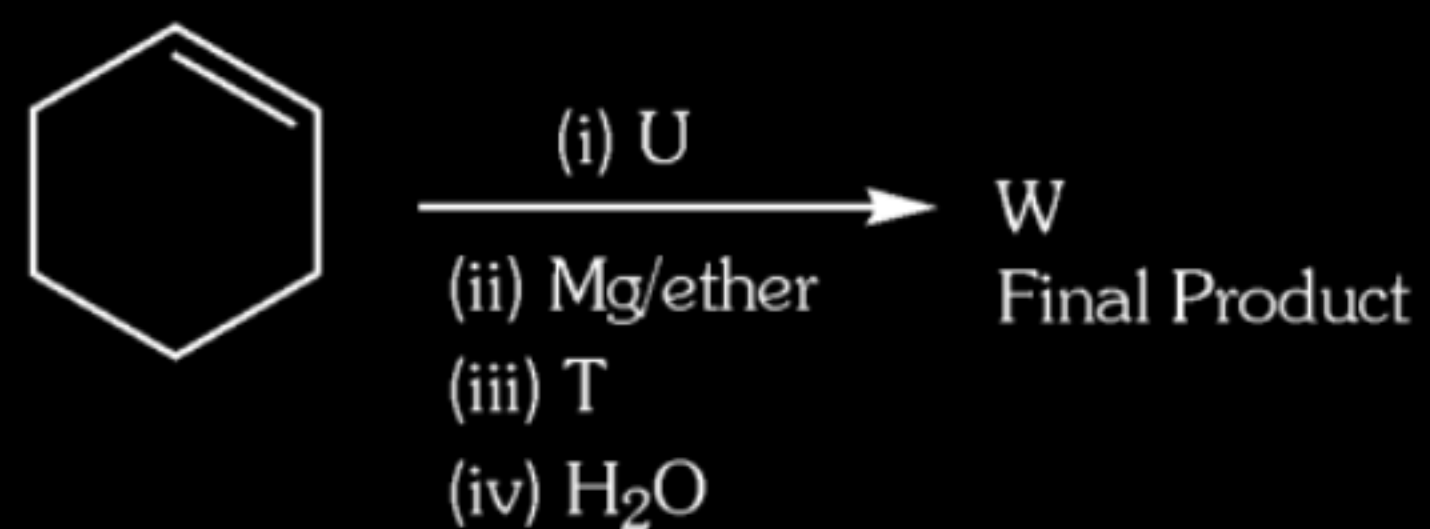
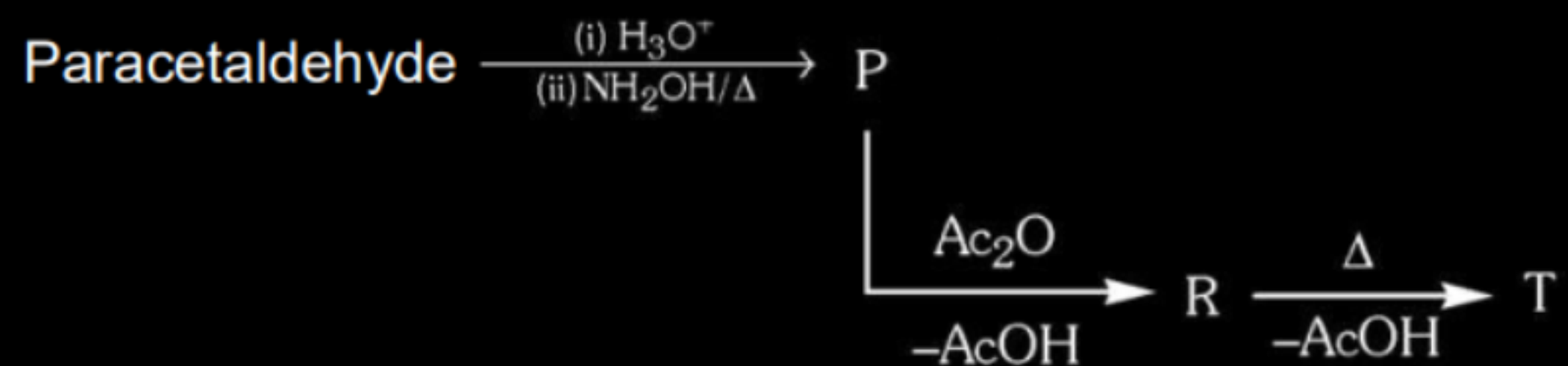
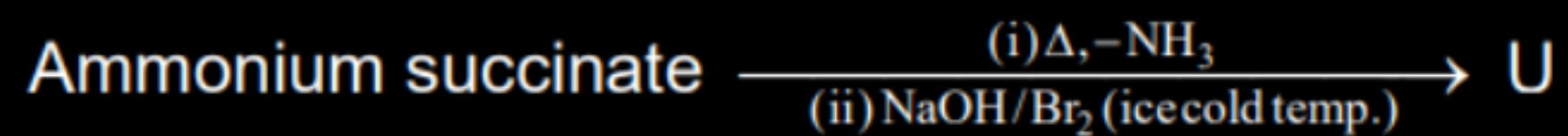
δ^-
MeMgCl



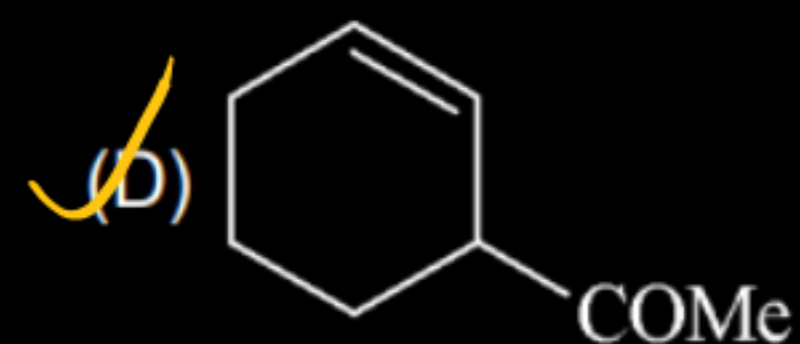
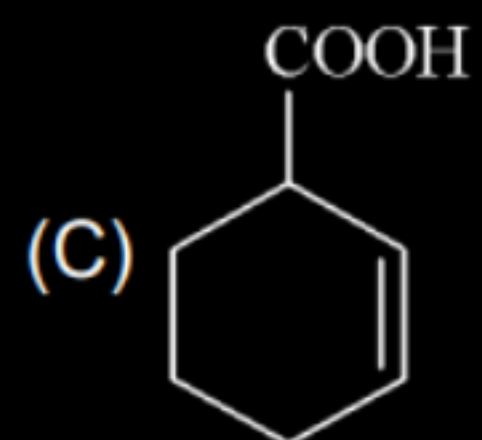
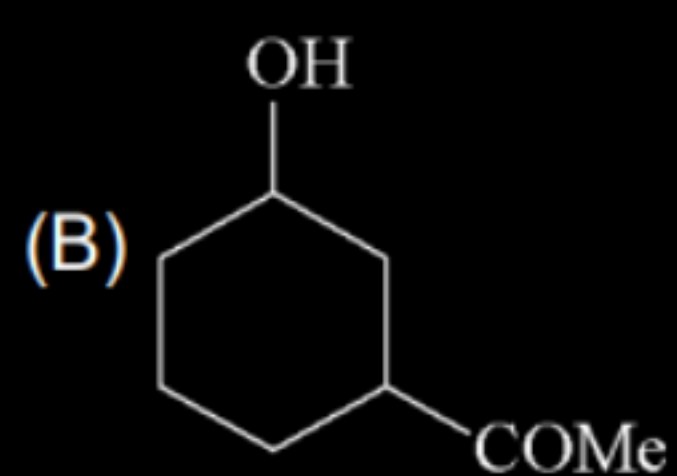
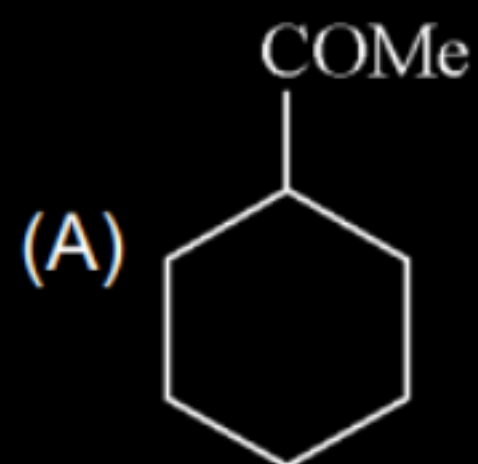


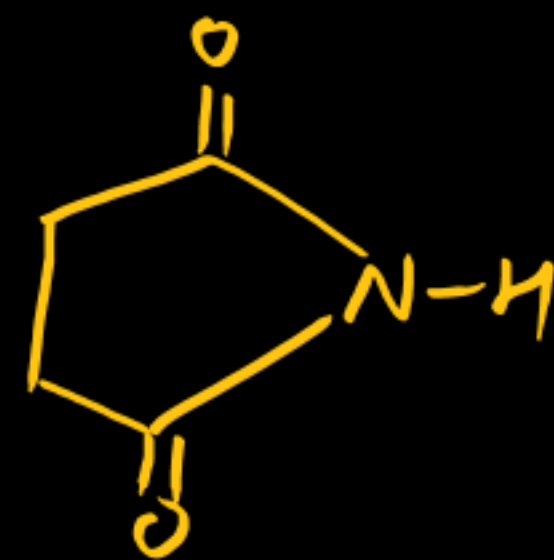
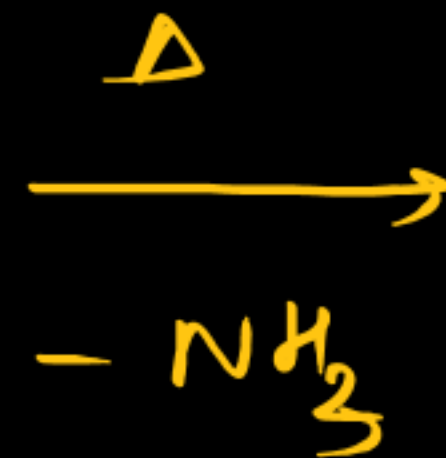
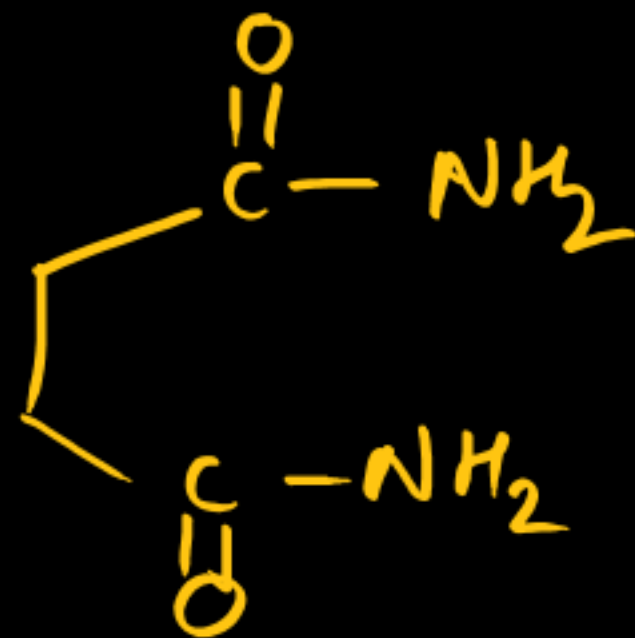
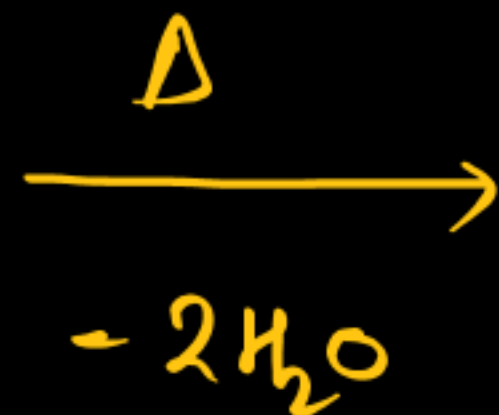
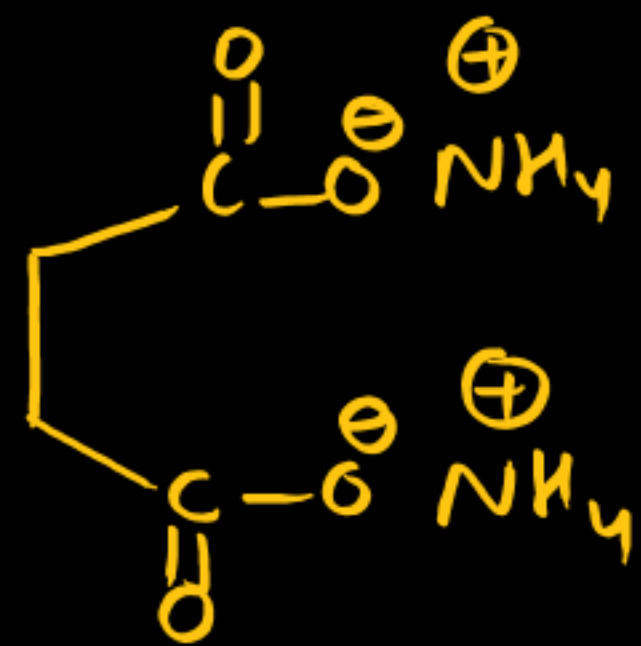


Δ - CO_2
 decarboxylation

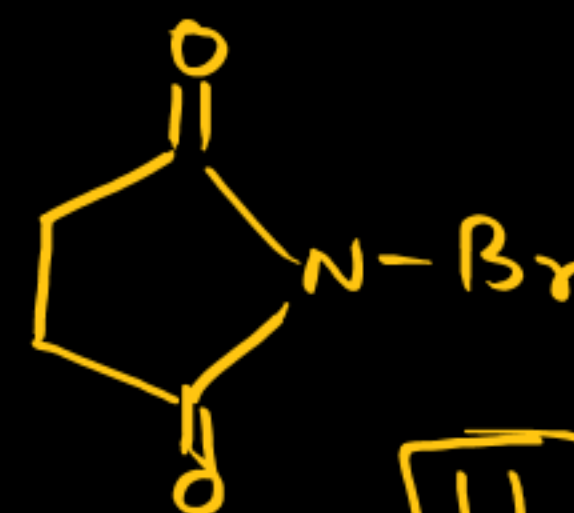


Identify final product 'W'.



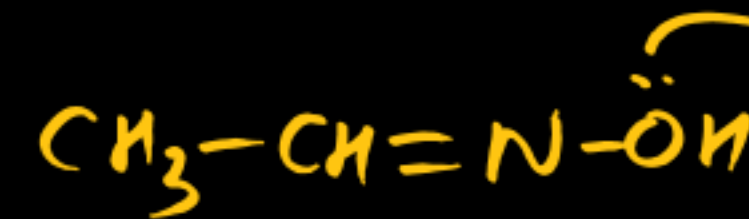
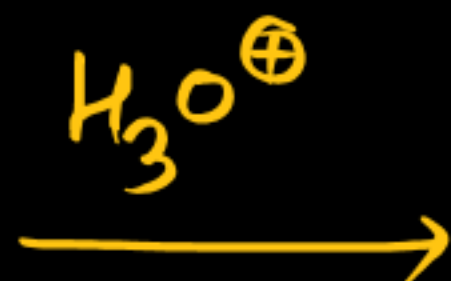
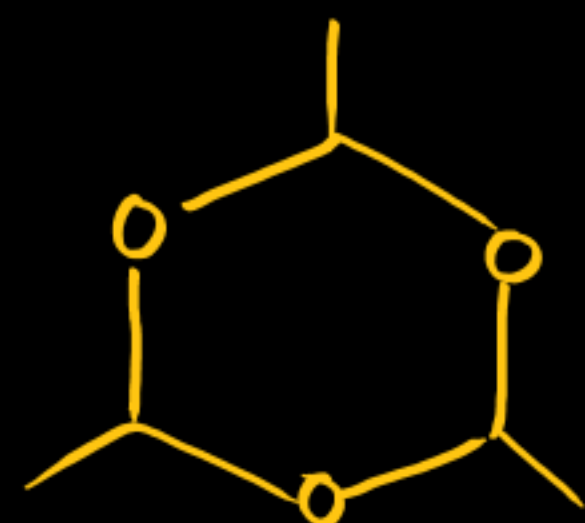


Succinamide

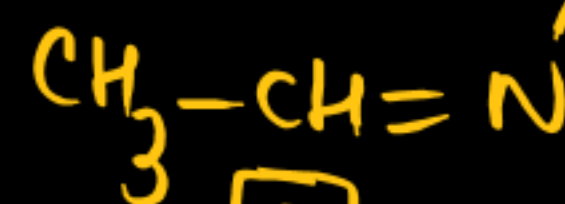
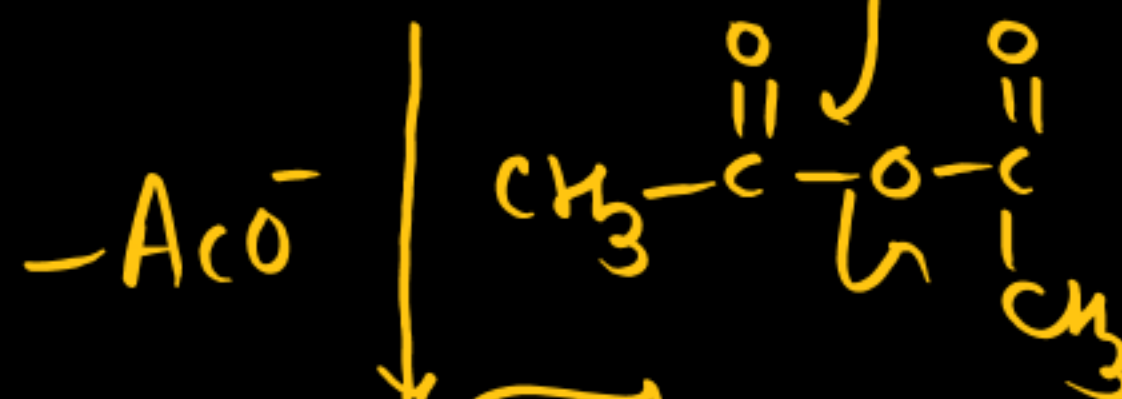


(NBS)

U



P Aldoxime

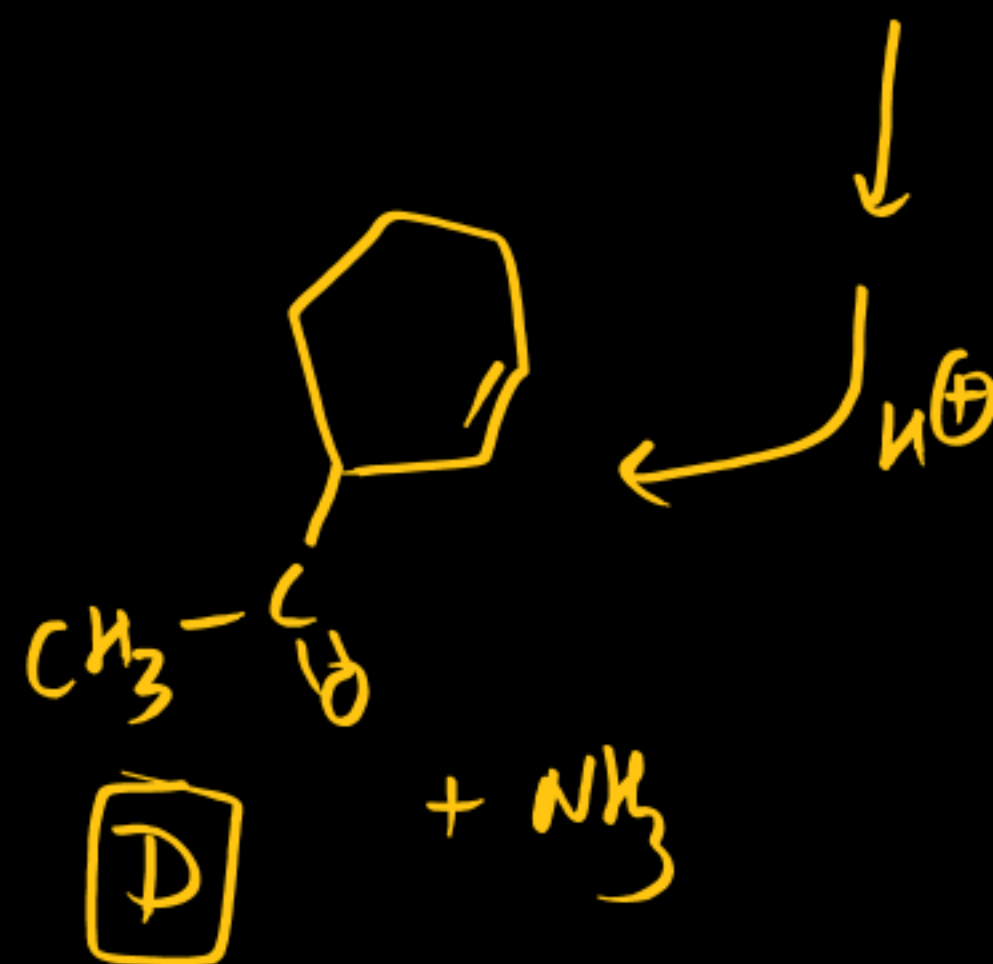
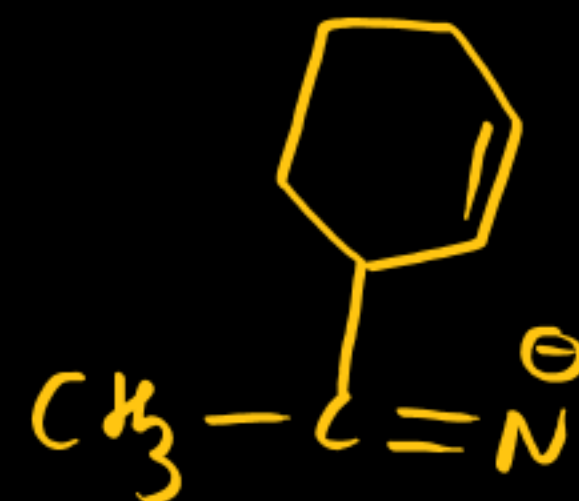
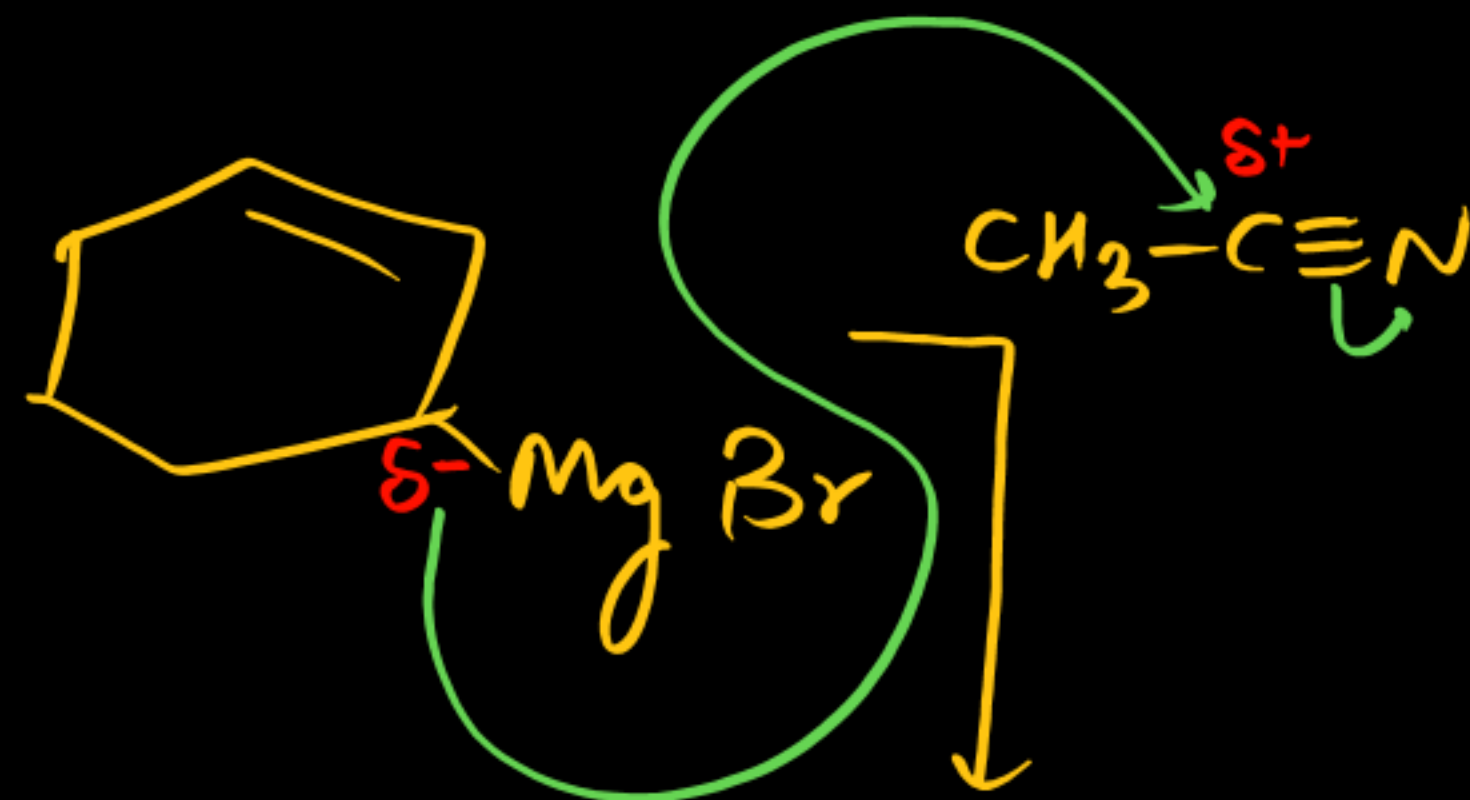
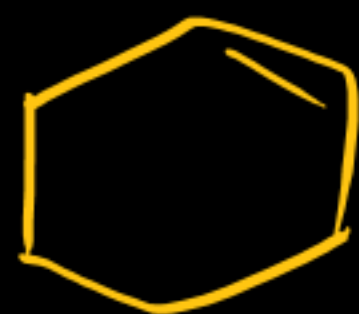


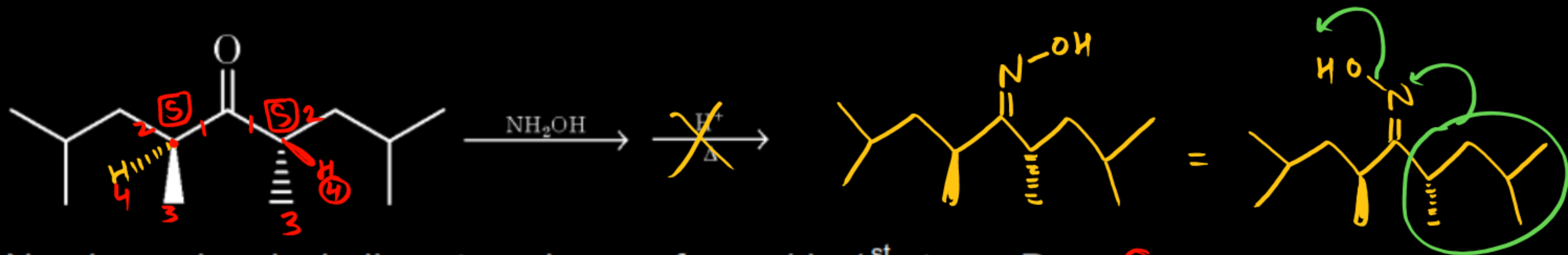
R

Backmann's Rxn



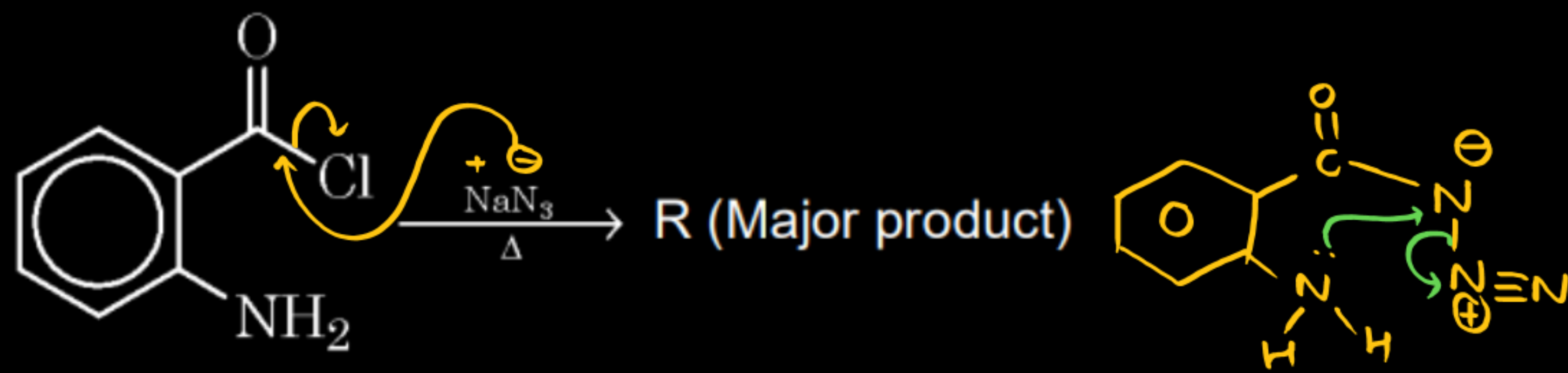
T





Number oxime including stereoisomer formed in 1st step = P = ①

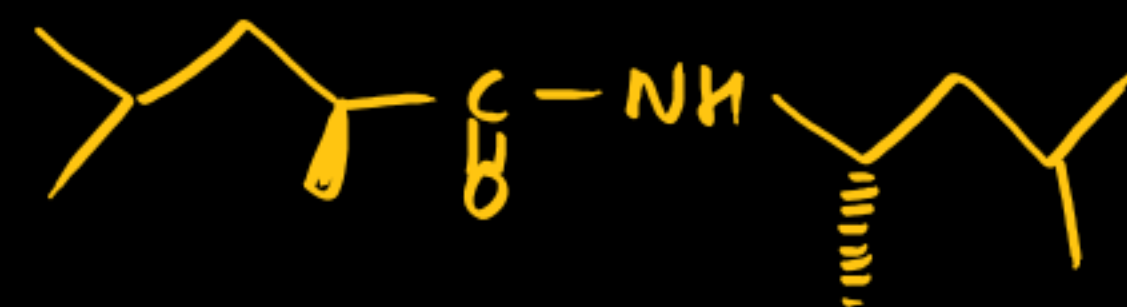
Number of amides including stereoisomer formed in 2nd step = Q = 1



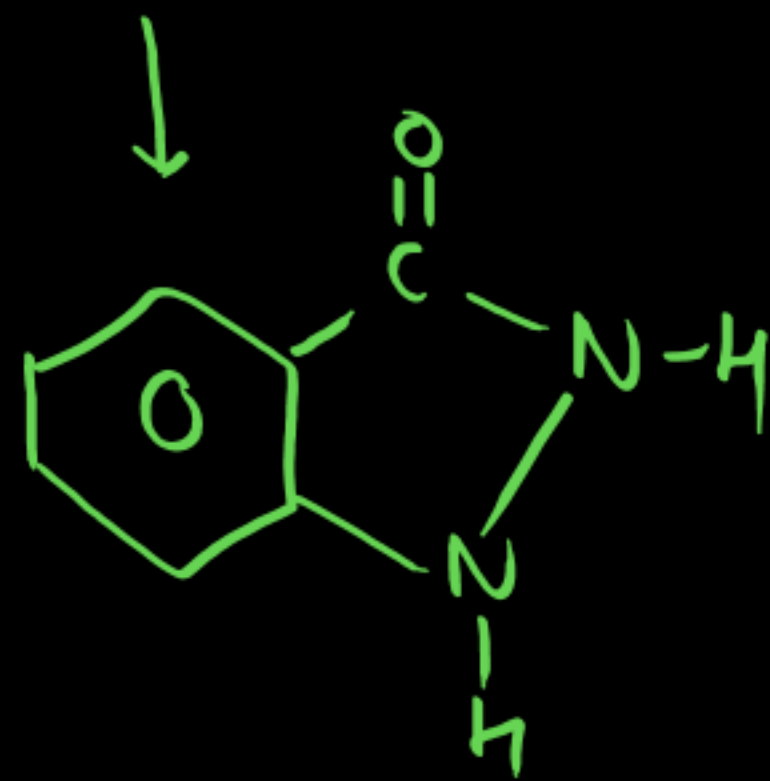
Number of 'N' atom in R in ring structure is 'S'. = 2

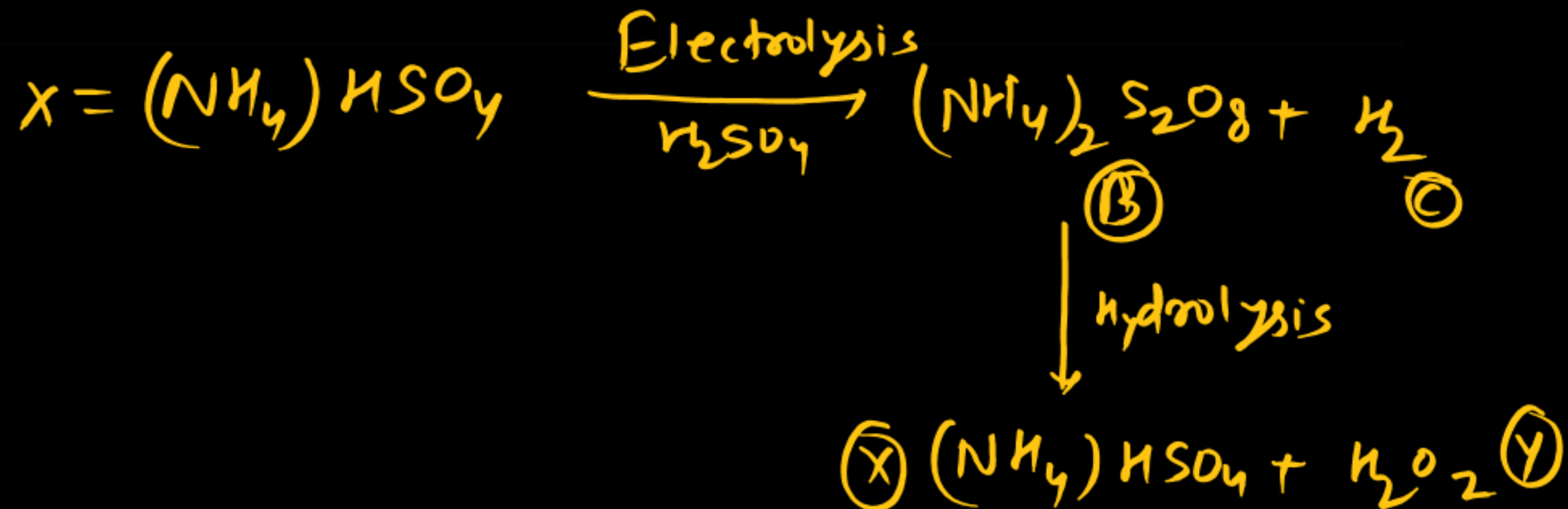
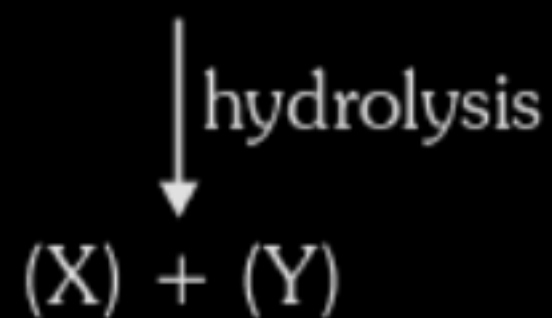
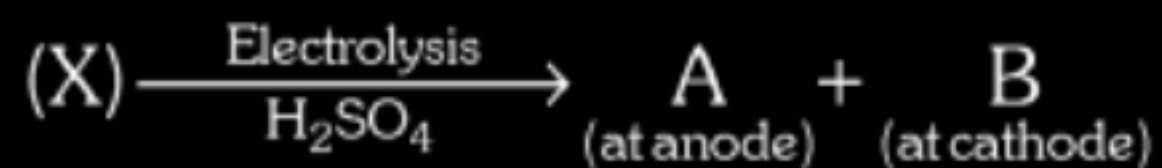
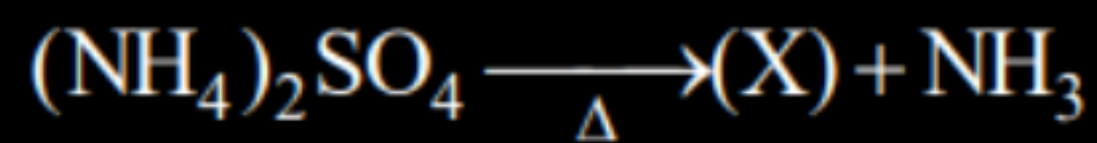
Give value of $\frac{P+Q}{S} = \frac{1+1}{2} = \boxed{1}$

↓ H^+ , Δ Beckmann's Rearrang -

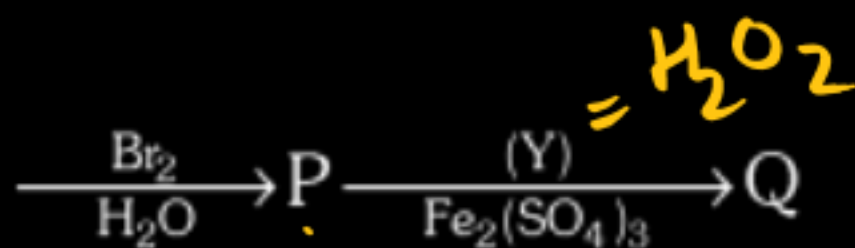
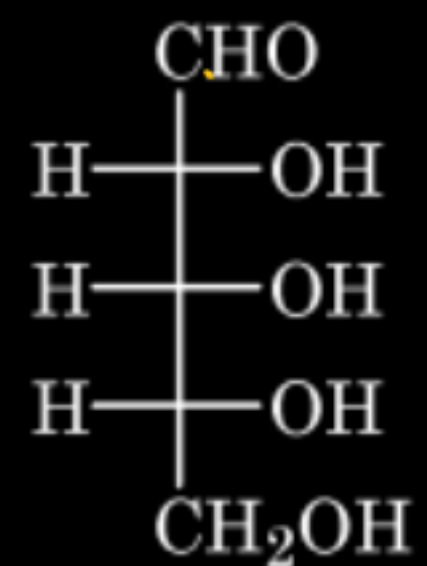


Retention of configuration





(X) + (Y) The structure of the major product Q formed in the following reaction sequence is:



Fenton's Reagent

